

UNITED STATES
SECURITIES AND EXCHANGE COMMISSION
Washington, D.C. 20549
FORM 10-K

(Mark One)

☒ ANNUAL REPORT PURSUANT TO SECTION 13 OR 15(d) OF THE SECURITIES EXCHANGE ACT OF 1934

For the fiscal year ended December 31, 2024
OR

☐ TRANSITION REPORT PURSUANT TO SECTION 13 OR 15(d) OF THE SECURITIES EXCHANGE ACT OF 1934

For the transition period from _____ to _____
Commission File Number: 001-34756

Tesla, Inc.

(Exact name of registrant as specified in its charter)

Texas
(State or other jurisdiction of
incorporation or organization)

91-2197729
(I.R.S. Employer
Identification No.)

1 Tesla Road
Austin, Texas
(Address of principal executive offices)

78725
(Zip Code)

(512) 516-8177
(Registrant's telephone number, including area code)

Securities registered pursuant to Section 12(b) of the Act:

Title of each class	Trading Symbol(s)	Name of each exchange on which registered
Common stock	TSLA	The Nasdaq Global Select Market

Securities registered pursuant to Section 12(g) of the Act:

None

Indicate by check mark whether the registrant is a well-known seasoned issuer, as defined in Rule 405 of the Securities Act. Yes ☐ No ☒

Indicate by check mark if the registrant is not required to file reports pursuant to Section 13 or 15(d) of the Act. Yes ☐ No ☒

Indicate by check mark whether the registrant (1) has filed all reports required to be filed by Section 13 or 15(d) of the Securities Exchange Act of 1934 ("Exchange Act") during the preceding 12 months (or for such shorter period that the registrant was required to file such reports), and (2) has been subject to such filing requirements for the past 90 days. Yes ☒ No ☐

Indicate by check mark whether the registrant has submitted electronically every Interactive Data File required to be submitted pursuant to Rule 405 of Regulation S-T (§232.405 of this chapter) during the preceding 12 months (or for such shorter period that the registrant was required to submit such files). Yes ☒ No ☐

Indicate by check mark whether the registrant is a large accelerated filer, an accelerated filer, a non-accelerated filer, a smaller reporting company, or an emerging growth company. See the definitions of "large accelerated filer," "accelerated filer," "smaller reporting company" and "emerging growth company" in Rule 12b-2 of the Exchange Act:

Large accelerated filer	☒	Accelerated filer	☐
Non-accelerated filer	☐	Smaller reporting company	☐
Emerging growth company	☐		

TESLA, INC.

ANNUAL REPORT ON FORM 10-K FOR THE YEAR ENDED DECEMBER 31, 2024

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Management Opportunities, Challenges and Uncertainties and 2025 Outlook

Automotive—Production

The following is a summary of the status of production of each of our announced vehicle models in production and under development, as of the date of this Annual Report on Form 10-K:

Region	Vehicle Model(s)	Production Status
California	Model S / Model X	Active
	Model 3 / Model Y	Active
Shanghai	Model 3 / Model Y	Active
Berlin	Model Y	Active
Texas	Model Y	Active
	Cybertruck	Active
	Cybercab	In development
Nevada	Tesla Semi	Pilot production
TBD	Roadster	In development

We are focused on growing our manufacturing capacity, which includes capacity for manufacturing newer vehicle models such as our Cybertruck, Tesla Semi and future vehicles utilizing aspects of our next generation platform, and ramping the production at our Gigafactories to their installed production capacities as well as increasing production rate and efficiency at our current factories. The next phase of production growth will depend on the continued ramp at our factories and be initiated by advances in autonomy and the introduction of new products, including those built on our next generation vehicle platform, as well as our ability to add to our available sources of battery cell supply by manufacturing our own cells that we are developing to have high-volume output, lower capital and production costs and longer range. Our goals are to improve vehicle performance, decrease production costs and increase affordability and customer awareness.

These plans are subject to uncertainties inherent in establishing and ramping manufacturing operations, which may be exacerbated by new product and manufacturing technologies we introduce, the number of concurrent international projects, any industry-wide component constraints, labor shortages and any future impact from events outside of our control. For example, during the first quarter of 2024, we experienced a sequential decline in production volumes partially caused by the early phase of the production ramp of the updated Model 3 at our Fremont factory, and factory shutdowns at Gigafactory Berlin-Brandenburg resulting from shipping diversions caused by the Red Sea conflict and an arson attack. In the first quarter of 2025, as we launch our New Model Y worldwide, we may similarly experience delays or declines in production volumes due to simultaneous manufacturing ramps in facilities on three continents. Moreover, we have set ambitious technological targets with our plans for battery cells as well as for iterative manufacturing and design improvements for our vehicles with each new factory.

Automotive—Demand, Sales, Deliveries and Infrastructure

Our cost reduction efforts, cost innovation strategies, and additional localized procurement and manufacturing are key to our vehicles' affordability and have allowed us to competitively price our vehicles. We will also continue to generate demand by improving our vehicles' performance and functionality, including through product offerings and features based on artificial intelligence such as Autopilot, FSD (Supervised), and other software, and delivering new vehicles and vehicle options, such as our launch of the updated Model 3 in 2024, and the New Model Y in the first quarter of 2025. In addition, we have been increasing awareness, and expanding our vehicle financing programs, including attractive leasing terms for our customers.

Furthermore, significant judgment is required in evaluating our tax positions. In the ordinary course of business, there are many transactions and calculations for which the ultimate tax settlement is uncertain. As a result, we recognize the effect of this uncertainty on our tax attributes or taxes payable based on our estimates of the eventual outcome. These effects are recognized when, despite our belief that our tax return positions are supportable, we believe that it is more likely than not that some of those positions may not be fully sustained upon review by tax authorities. We are required to file income tax returns in the U.S. and various foreign jurisdictions, which requires us to interpret the applicable tax laws and regulations in effect in such jurisdictions. Such returns are subject to audit by the various federal, state and foreign taxing authorities, who may disagree with respect to our tax positions. We believe that our consideration is adequate for all open audit years based on our assessment of many factors, including past experience and interpretations of tax law. We review and update our estimates in light of changing facts and circumstances, such as the closing of a tax audit, the lapse of a statute of limitations or a change in estimate. To the extent that the final tax outcome of these matters differs from our expectations, such differences may impact income tax expense in the period in which such determination is made.

Results of Operations

Revenues

(Dollars in millions)	Year Ended December 31,			2024 vs. 2023 Change		2023 vs. 2022 Change	
	2024	2023	2022	\$	%	\$	%
Automotive sales	\$ 72,480	\$ 78,509	\$ 67,210	\$ (6,029)	(8)%	\$ 11,299	17 %
Automotive regulatory credits	2,763	1,790	1,776	973	54 %	14	1 %
Automotive leasing	1,827	2,120	2,476	(293)	(14)%	(356)	(14)%
Total automotive revenues	77,070	82,419	71,462	(5,349)	(6)%	10,957	15 %
Services and other	10,534	8,319	6,091	2,215	27 %	2,228	37 %
Total automotive & services and other segment revenue	87,604	90,738	77,553	(3,134)	(3)%	13,185	17 %
Energy generation and storage segment revenue	10,086	6,035	3,909	4,051	67 %	2,126	54 %
Total revenues	\$ 97,690	\$ 96,773	\$ 81,462	\$ 917	1 %	\$ 15,311	19 %

Automotive & Services and Other Segment

Automotive sales revenue includes revenues related to cash and financing deliveries of new vehicles, including access to our FSD (Supervised) features and their ongoing maintenance, internet connectivity, free Supercharging programs and over-the-air software updates. These deliveries are vehicles that are not subject to lease accounting.

Automotive regulatory credits includes sales of regulatory credits to other automotive manufacturers. Our revenue from automotive regulatory credits is directly related to our new vehicle production, sales and pricing negotiated with our customers. We monetize them proactively as new vehicles are sold based on standing arrangements with buyers of such credits, typically as close as possible to the production and delivery of the vehicle or changes in regulation impacting the credits.

Automotive leasing revenue includes the amortization of revenue for vehicles under direct operating lease agreements. Additionally, automotive leasing revenue includes direct sales-type leasing programs where we recognize all revenue associated with the sales-type lease upon delivery to the customer.

Services and other revenue consists of sales of used vehicles, non-warranty maintenance services and collision, part sales, paid Supercharging, insurance services revenue and retail merchandise sales.

2024 compared to 2023

Automotive sales revenue decreased \$6.03 billion, or 8%, in the year ended December 31, 2024 as compared to the year ended December 31, 2023, primarily due to lower average selling price on our vehicles driven by overall price reductions and attractive financing options provided in 2024 as well as mix. Additionally, there was a decrease of approximately 22,000 combined Model 3 and Model Y cash deliveries. The decreases were partially offset by an increase of approximately 19,000 deliveries of other models primarily due to our production ramp of Cybertruck. Additionally, we recognized \$596 million of FSD (Supervised) revenue due to release of certain features in 2024.

Automotive regulatory credits revenue increased \$973 million, or 54%, in the year ended December 31, 2024 as compared to the year ended December 31, 2023, driven by demand for credits in North America as other automobile manufacturers scale back on their battery electric vehicle plans.

Automotive leasing revenue decreased \$293 million, or 14%, in the year ended December 31, 2024 as compared to the year ended December 31, 2023, primarily due to lower direct sales-type leasing deliveries as we have shifted towards providing leasing options through commercial banking partner programs that allow for us to recognize upfront revenue in automotive sales and a decrease in lease buyouts.

Services and other revenue increased \$2.22 billion, or 27%, in the year ended December 31, 2024 as compared to the year ended December 31, 2023, primarily due to increases in sales of used vehicles, non-warranty maintenance services and collision revenue, paid Supercharging revenue, insurance services revenue and part sales revenue.

Energy Generation and Storage Segment

Energy generation and storage revenue includes sales and leasing of solar energy generation and energy storage products, financing of solar energy generation products, services related to such products and sales of solar energy systems incentives.

2024 compared to 2023

Energy generation and storage revenue increased \$4.05 billion, or 67%, in the year ended December 31, 2024 as compared to the year ended December 31, 2023, primarily due to a 16.7 GWh increase in Megapack and Powerwall deployments compared to the prior year.

Cost of Revenues and Gross Margin

(Dollars in millions)	Year Ended December 31,			2024 vs. 2023 Change		2023 vs. 2022 Change	
	2024	2023	2022	\$	%	\$	%
Cost of revenues							
Automotive sales	\$ 61,870	\$ 65,121	\$ 49,599	\$ (3,251)	(5)%	\$ 15,522	31 %
Automotive leasing	1,003	1,268	1,509	(265)	(21)%	(241)	(16)%
Total automotive cost of revenues	62,873	66,389	51,108	(3,516)	(5)%	15,281	30 %
Services and other	9,921	7,830	5,880	2,091	27 %	1,950	33 %
Total automotive & services and other segment cost of revenues	72,794	74,219	56,988	(1,425)	(2)%	17,231	30 %
Energy generation and storage segment	7,446	4,894	3,621	2,552	52 %	1,273	35 %
Total cost of revenues	\$ 80,240	\$ 79,113	\$ 60,609	\$ 1,127	1 %	\$ 18,504	31 %
Gross profit total automotive							
	\$ 14,197	\$ 16,030	\$ 20,354				
Gross margin total automotive							
	18.4 %	19.4 %	28.5 %				
Gross profit total automotive & services and other segment							
	\$ 14,810	\$ 16,519	\$ 20,565				
Gross margin total automotive & services and other segment							
	16.9 %	18.2 %	26.5 %				
Gross profit energy generation and storage segment							
	\$ 2,640	\$ 1,141	\$ 288				
Gross margin energy generation and storage segment							
	26.2 %	18.9 %	7.4 %				
Total gross profit							
	\$ 17,450	\$ 17,660	\$ 20,853				
Total gross margin							
	17.9 %	18.2 %	25.6 %				

Gross margin for energy generation and storage increased from 18.9% to 26.2% in the year ended December 31, 2024 as compared to the year ended December 31, 2023, primarily due to margin improvements for our energy storage products driven by cost reductions, including benefits from IRA manufacturing credits, and a higher proportion of our storage business, which operated at a higher gross margin, within the segment as compared to the prior periods.

Research and Development Expense

(Dollars in millions)	Year Ended December 31,			2024 vs. 2023 Change		2023 vs. 2022 Change	
	2024	2023	2022	\$	%	\$	%
Research and development	\$ 4,540	\$ 3,969	\$ 3,075	\$ 571	14 %	\$ 894	29 %
As a percentage of revenues	5 %	4 %	4 %				

Research and development (“R&D”) expenses consist primarily of personnel costs for our teams in engineering and research, manufacturing engineering and manufacturing test organizations, prototyping expense, contract and professional services and amortized equipment expense.

R&D expenses increased \$571 million, or 14%, in the year ended December 31, 2024 as compared to the year ended December 31, 2023. The overall increases were primarily driven by additional costs year over year related to AI programs. R&D expenses as a percentage of revenue increased from 4% to 5% in the year ended December 31, 2024 as compared to the year ended December 31, 2023 as we continue to expand our product roadmap and technologies.

Selling, General and Administrative Expense

(Dollars in millions)	Year Ended December 31,			2024 vs. 2023 Change		2023 vs. 2022 Change	
	2024	2023	2022	\$	%	\$	%
Selling, general and administrative	\$ 5,150	\$ 4,800	\$ 3,946	\$ 350	7 %	\$ 854	22 %
As a percentage of revenues	5 %	5 %	5 %				

Selling, general and administrative (“SG&A”) expenses generally consist of personnel and facilities costs related to our stores, marketing, sales, executive, finance, human resources, information technology and legal organizations, as well as fees for professional and contract services and litigation settlements.

SG&A expenses increased \$350 million, or 7%, in the year ended December 31, 2024 as compared to the year ended December 31, 2023 driven by a \$171 million increase in facilities related expenses, a \$115 million increase in employee and labor costs, including professional services and a \$57 million increase in marketing expenses.

Restructuring and Other

(Dollars in millions)	Year Ended December 31,			2024 vs. 2023 Change		2023 vs. 2022 Change	
	2024	2023	2022	\$	%	\$	%
Restructuring and other	\$ 684	\$ —	\$ 176	\$ 684	Not meaningful	\$ (176)	(100)%

In the second quarter of 2024, we initiated and substantially completed certain restructuring actions to reduce costs and improve efficiency. As a result, we recognized \$583 million of employee termination expenses in Restructuring and other in our consolidated income statement. These expenses were substantially paid with an immaterial accrual remaining in Accrued liabilities and other in our consolidated balance sheet as of December 31, 2024.

Interest Income

(Dollars in millions)	Year Ended December 31,			2024 vs. 2023 Change		2023 vs. 2022 Change	
	2024	2023	2022	\$	%	\$	%
Interest income	\$ 1,569	\$ 1,066	\$ 297	\$ 503	47 %	\$ 769	259 %

Interest income increased \$503 million, or 47%, in the year ended December 31, 2024 as compared to the year ended December 31, 2023, primarily due to higher interest earned on our cash and cash equivalents and short-term investments compared to the prior year due to an increase in our portfolio balance and a higher weighted average interest rate.

Other Income (Expense), Net

(Dollars in millions)	Year Ended December 31,			2024 vs. 2023 Change		2023 vs. 2022 Change	
	2024	2023	2022	\$	%	\$	%
Other income (expense), net	\$ 695	\$ 172	\$ (43)	\$ 523	304%	\$ 215	Not meaningful

Other income (expense), net, consists of foreign exchange gains and losses related to our foreign currency-denominated monetary assets and liabilities and fair value remeasurements of our digital assets following the adoption of ASU 2023-08 effective January 1, 2024. See Note 2, *Summary of Significant Accounting Policies*, to the consolidated financial statements included elsewhere in this Annual Report on Form 10-K for further details.

Other income (expense), net, changed favorably by \$523 million in the year ended December 31, 2024 as compared to the year ended December 31, 2023 primarily due to remeasurement of our bitcoin digital assets to fair value in 2024 (see above), partially offset by unfavorable fluctuations in foreign currency exchange rates on our intercompany balances. As our intercompany balances are significant in nature and we do not typically hedge foreign currency risk, we can experience significant fluctuations in foreign currency exchange rate gains and losses from period to period.

Provision for (Benefit from) Income Taxes

(Dollars in millions)	Year Ended December 31,			2024 vs. 2023 Change		2023 vs. 2022 Change	
	2024	2023	2022	\$	%	\$	%
Provision for (benefit from) income taxes	\$ 1,837	\$ (5,001)	\$ 1,132	\$ 6,838	Not meaningful	\$ (6,133)	Not meaningful
Effective tax rate	20 %	(50)%	8 %				

Our provision for (benefit from) income taxes changed by \$6.84 billion in the year ended December 31, 2024 as compared to the year ended December 31, 2023. Our effective tax rate changed to an expense of 20% in the year ended December 31, 2024 from a benefit of 50% in the year ended December 31, 2023. These changes are primarily due to the impact of releasing the valuation allowance on our U.S. deferred tax assets in the fourth quarter of 2023.

We monitor the realizability of our deferred tax assets taking into account all relevant factors at each reporting period. In the fourth quarter of 2023, based on the relevant weight of positive and negative evidence, including the amount of our taxable income in recent years which was objective and verifiable, and consideration of our expected future taxable earnings, we concluded that it is more likely than not that most of our U.S. federal and certain state deferred tax assets are realizable and released \$6.54 billion of our valuation allowance. As of December 31, 2024, we continue to conclude that it is more likely than not that our deferred tax assets are realizable, except for our California deferred tax assets, U.S. foreign tax credits and certain foreign operating losses.

See Note 13, *Income Taxes*, to the consolidated financial statements included elsewhere in this Annual Report on Form 10-K for further details.

Liquidity and Capital Resources

We expect to continue to generate net positive operating cash flow. The cash we generate from our core operations enables us to fund ongoing operations and production, our research and development projects for new products and technologies including our proprietary battery cells, additional manufacturing ramps at existing manufacturing facilities, the construction of future factories, and the continued expansion of our retail and service locations, body shops, Mobile Service fleet, Supercharger, including to support NACS, energy product installation capabilities and autonomy and other artificial intelligence enabled products.

In addition, because a large portion of our future expenditures will be to fund our growth, we expect that if needed we will be able to adjust our capital and operating expenditures by operating segment. For example, if our near-term manufacturing operations decrease in scale or ramp more slowly than expected, including due to global economic or business conditions, we may choose to correspondingly slow the pace of our capital expenditures. Finally, we continually evaluate our cash needs and may decide it is best to raise additional capital or seek alternative financing sources to fund the rapid growth of our business, including through drawdowns on existing or new debt facilities or financing funds. Conversely, we may also from time to time determine that it is in our best interests to voluntarily repay certain indebtedness early.

Accordingly, we believe that our current sources of funds will provide us with adequate liquidity during the 12-month period following December 31, 2024, as well as in the long-term.

See the sections below for more details regarding the material requirements for cash in our business and our sources of liquidity to meet such needs.

Material Cash Requirements

From time to time in the ordinary course of business, we enter into agreements with vendors for the purchase of components and raw materials to be used in the manufacture of our products. However, due to contractual terms, variability in the precise growth curves of our development and production ramps, and opportunities to renegotiate pricing, we generally do not have binding and enforceable purchase orders under such contracts beyond the short-term, and the timing and magnitude of purchase orders beyond such period is difficult to accurately project.

As discussed in and subject to the considerations referenced in Part II, Item 7, Management's Discussion and Analysis of Financial Condition and Results of Operations—Management Opportunities, Challenges and Uncertainties and 2025 Outlook—Cash Flow and Capital Expenditure Trends in this Annual Report on Form 10-K, we currently expect our capital expenditures to support our projects globally to exceed \$11.00 billion in 2025 and in each of the following two fiscal years.

As of December 31, 2024, we and our subsidiaries had outstanding \$7.91 billion in aggregate principal amount of indebtedness, of which \$2.35 billion is current. As of December 31, 2024, our total minimum lease payments was \$7.05 billion, of which \$1.19 billion is due in the succeeding 12 months. For details regarding our indebtedness and lease obligations, refer to Note 10, *Debt*, and Note 11, *Leases*, to the consolidated financial statements included elsewhere in this Annual Report on Form 10-K.

Sources and Conditions of Liquidity

Our sources to fund our material cash requirements are predominantly from our deliveries and servicing of new and used vehicles, sales and installations of our energy storage products, interest income, and proceeds from debt facilities and equity offerings, when applicable.

As of December 31, 2024, we had \$16.14 billion and \$20.42 billion of cash and cash equivalents and short-term investments, respectively. Balances held in foreign currencies had a U.S. dollar equivalent of \$2.90 billion and consisted primarily of euros and Chinese yuan. We had \$5.00 billion of unused committed credit amounts as of December 31, 2024. For details regarding our indebtedness, refer to Note 10, *Debt*, to the consolidated financial statements included elsewhere in this Annual Report on Form 10-K.

We continue adapting our strategy to meet our liquidity and risk objectives, such as investing in U.S. government securities and other investments, invest in autonomy, do more vertical integration, expand our product roadmap and provide financing options to our customers.

Summary of Cash Flows

(Dollars in millions)	Year Ended December 31,		
	2024	2023	2022
Net cash provided by operating activities	\$ 14,923	\$ 13,256	\$ 14,724
Net cash used in investing activities	\$ (18,787)	\$ (15,584)	\$ (11,973)
Net cash provided by (used in) financing activities	\$ 3,853	\$ 2,589	\$ (3,527)

Cash Flows from Operating Activities

Our cash flows from operating activities are significantly affected by our cash investments to support the growth of our business in areas such as research and development and selling, general and administrative and working capital. Our operating cash inflows include cash from vehicle sales and related servicing, customer lease and financing payments, customer deposits, cash from sales of regulatory credits and energy generation and storage products, and interest income on our cash and investments portfolio. These cash inflows are offset by our payments to suppliers for production materials and parts used in our manufacturing process, operating expenses, operating lease payments and interest payments on our financings.

Tesla, Inc.
Consolidated Balance Sheets
(in millions, except per share data)

	December 31, 2024	December 31, 2023
Assets		
Current assets		
Cash and cash equivalents	\$ 16,139	\$ 16,398
Short-term investments	20,424	12,696
Accounts receivable, net	4,418	3,508
Inventory	12,017	13,626
Prepaid expenses and other current assets	5,362	3,388
Total current assets	58,360	49,616
Operating lease vehicles, net	5,581	5,989
Solar energy systems, net	4,924	5,229
Property, plant and equipment, net	35,836	29,725
Operating lease right-of-use assets	5,160	4,180
Digital assets, net	1,076	184
Intangible assets, net	150	178
Goodwill	244	253
Deferred tax assets	6,524	6,733
Other non-current assets	4,215	4,531
Total assets	\$ 122,070	\$ 106,618
Liabilities		
Current liabilities		
Accounts payable	\$ 12,474	\$ 14,431
Accrued liabilities and other	10,723	9,080
Deferred revenue	3,168	2,864
Current portion of debt and finance leases	2,456	2,373
Total current liabilities	28,821	28,748
Debt and finance leases, net of current portion	5,757	2,857
Deferred revenue, net of current portion	3,317	3,251
Other long-term liabilities	10,495	8,153
Total liabilities	48,390	43,009
Commitments and contingencies (Note 14)		
Redeemable noncontrolling interests in subsidiaries	63	242
Equity		
Stockholders' equity		
Preferred stock; \$0.001 par value; 100 shares authorized; no shares issued and outstanding	—	—
Common stock; \$0.001 par value; 6,000 shares authorized; 3,216 and 3,185 shares issued and outstanding as of December 31, 2024 and December 31, 2023, respectively	3	3
Additional paid-in capital	38,371	34,892
Accumulated other comprehensive loss	(670)	(143)
Retained earnings	35,209	27,882
Total stockholders' equity	72,913	62,634
Noncontrolling interests in subsidiaries	704	733
Total liabilities and equity	\$ 122,070	\$ 106,618

The accompanying notes are an integral part of these consolidated financial statements.

Tesla, Inc.
Consolidated Statements of Operations
(in millions, except per share data)

	Year Ended December 31,		
	2024	2023	2022
Revenues			
Automotive sales	\$ 72,480	\$ 78,509	\$ 67,210
Automotive regulatory credits	2,763	1,790	1,776
Automotive leasing	1,827	2,120	2,476
Total automotive revenues	77,070	82,419	71,462
Energy generation and storage	10,086	6,035	3,909
Services and other	10,534	8,319	6,091
Total revenues	97,690	96,773	81,462
Cost of revenues			
Automotive sales	61,870	65,121	49,599
Automotive leasing	1,003	1,268	1,509
Total automotive cost of revenues	62,873	66,389	51,108
Energy generation and storage	7,446	4,894	3,621
Services and other	9,921	7,830	5,880
Total cost of revenues	80,240	79,113	60,609
Gross profit	17,450	17,660	20,853
Operating expenses			
Research and development	4,540	3,969	3,075
Selling, general and administrative	5,150	4,800	3,946
Restructuring and other	684	—	176
Total operating expenses	10,374	8,769	7,197
Income from operations	7,076	8,891	13,656
Interest income	1,569	1,066	297
Interest expense	(350)	(156)	(191)
Other income (expense), net	695	172	(43)
Income before income taxes	8,990	9,973	13,719
Provision for (benefit from) income taxes	1,837	(5,001)	1,132
Net income	7,153	14,974	12,587
Net income (loss) attributable to noncontrolling interests and redeemable noncontrolling interests in subsidiaries	62	(23)	31
Net income attributable to common stockholders	\$ 7,091	\$ 14,997	\$ 12,556
Net income per share of common stock attributable to common stockholders			
Basic	\$ 2.23	\$ 4.73	\$ 4.02
Diluted	\$ 2.04	\$ 4.30	\$ 3.62
Weighted average shares used in computing net income per share of common stock			
Basic	3,197	3,174	3,130
Diluted	3,498	3,485	3,475

The accompanying notes are an integral part of these consolidated financial statements.

Tesla, Inc.
Consolidated Statements of Comprehensive Income
(in millions)

	Year Ended December 31,		
	2024	2023	2022
Net income	\$ 7,153	\$ 14,974	\$ 12,587
Other comprehensive income (loss):			
Foreign currency translation adjustment	(539)	198	(392)
Unrealized net gain (loss) on investments, net of tax	12	16	(23)
Net loss realized and included in net income	—	4	—
Comprehensive income	6,626	15,192	12,172
Less: Comprehensive income (loss) attributable to noncontrolling interests and redeemable noncontrolling interests in subsidiaries	62	(23)	31
Comprehensive income attributable to common stockholders	\$ 6,564	\$ 15,215	\$ 12,141

The accompanying notes are an integral part of these consolidated financial statements.

Tesla, Inc.

Consolidated Statements of Redeemable Noncontrolling Interests and Equity
(in millions)

	Redeemable Noncontrolling Interests	Common Stock		Additional Paid-In Capital	Accumulated Other Comprehensive Income (Loss)	Retained Earnings	Total Stockholders' Equity	Noncontrolling Interests in Subsidiaries	Total Equity
		Shares	Amount						
Balance as of December 31, 2021	\$ 568	3,100	\$ 3	\$ 29,803	\$ 54	\$ 329	\$ 30,189	\$ 826	\$ 31,015
Settlements of warrants	—	37	—	—	—	—	—	—	—
Issuance of common stock for equity incentive awards	—	27	—	541	—	—	541	—	541
Stock-based compensation	—	—	—	1,806	—	—	1,806	—	1,806
Distributions to noncontrolling interests	(46)	—	—	—	—	—	—	(113)	(113)
Buy-outs of noncontrolling interests	(11)	—	—	27	—	—	27	(61)	(34)
Net (loss) income	(102)	—	—	—	—	12,556	12,556	133	12,689
Other comprehensive loss	—	—	—	—	(415)	—	(415)	—	(415)
Balance as of December 31, 2022	\$ 409	3,164	\$ 3	\$ 32,177	\$ (361)	\$ 12,885	\$ 44,704	\$ 785	\$ 45,489
Issuance of common stock for equity incentive awards	—	21	—	700	—	—	700	—	700
Stock-based compensation	—	—	—	2,013	—	—	2,013	—	2,013
Distributions to noncontrolling interests	(32)	—	—	—	—	—	—	(108)	(108)
Buy-outs of noncontrolling interests	(39)	—	—	2	—	—	2	(17)	(15)
Net (loss) income	(96)	—	—	—	—	14,997	14,997	73	15,070
Other comprehensive income	—	—	—	—	218	—	218	—	218
Balance as of December 31, 2023	\$ 242	3,185	\$ 3	\$ 34,892	\$ (143)	\$ 27,882	\$ 62,634	\$ 733	\$ 63,367
Adjustments for prior periods from adopting ASU 2023-08, net of tax	—	—	—	—	—	236	236	—	236
Settlements of warrants	—	11	—	—	—	—	—	—	—
Issuance of common stock for equity incentive awards	—	20	—	1,241	—	—	1,241	—	1,241
Stock-based compensation	—	—	—	2,199	—	—	2,199	—	2,199
Distributions to noncontrolling interests	(13)	—	—	—	—	—	—	(85)	(85)
Buy-outs of noncontrolling interests	(172)	—	—	39	—	—	39	—	39
Net income	6	—	—	—	—	7,091	7,091	56	7,147
Other comprehensive loss	—	—	—	—	(527)	—	(527)	—	(527)
Balance as of December 31, 2024	\$ 63	3,216	\$ 3	\$ 38,371	\$ (670)	\$ 35,209	\$ 72,913	\$ 704	\$ 73,617

The accompanying notes are an integral part of these consolidated financial statements.

Tesla, Inc.
Consolidated Statements of Cash Flows
(in millions)

	Year Ended December 31,		
	2024	2023	2022
Cash Flows from Operating Activities			
Net income	\$ 7,153	\$ 14,974	\$ 12,587
Adjustments to reconcile net income to net cash provided by operating activities:			
Depreciation, amortization and impairment	5,368	4,667	3,747
Stock-based compensation	1,999	1,812	1,560
Inventory and purchase commitments write-downs	335	463	177
Foreign currency transaction net unrealized (gain) loss	(73)	(144)	81
Deferred income taxes	477	(6,349)	(196)
Non-cash interest and other operating activities	172	81	340
Digital assets (gain) loss, net	(589)	—	140
Changes in operating assets and liabilities:			
Accounts receivable	(1,083)	(586)	(1,124)
Inventory	937	(1,195)	(6,465)
Operating lease vehicles	(590)	(1,952)	(1,570)
Prepaid expenses and other assets	(3,273)	(2,652)	(3,713)
Accounts payable, accrued and other liabilities	3,588	2,605	8,029
Deferred revenue	502	1,532	1,131
Net cash provided by operating activities	14,923	13,256	14,724
Cash Flows from Investing Activities			
Purchases of property and equipment excluding finance leases, net of sales	(11,339)	(8,898)	(7,158)
Purchases of solar energy systems, net of sales	(3)	(1)	(5)
Proceeds from sales of digital assets	—	—	936
Purchase of intangible assets	—	—	(9)
Purchases of investments	(35,955)	(19,112)	(5,835)
Proceeds from maturities of investments	28,310	12,353	22
Proceeds from sales of investments	200	138	—
Receipt of government grants	—	—	76
Business combinations, net of cash acquired	—	(64)	—
Net cash used in investing activities	(18,787)	(15,584)	(11,973)
Cash Flows from Financing Activities			
Proceeds from issuances of debt	5,744	3,931	—
Repayments of debt	(2,500)	(1,351)	(3,364)
Proceeds from exercises of stock options and other stock issuances	1,241	700	541
Principal payments on finance leases	(381)	(464)	(502)
Debt issuance costs	(14)	(29)	—
Distributions paid to noncontrolling interests in subsidiaries	(104)	(144)	(157)
Payments for buy-outs of noncontrolling interests in subsidiaries	(133)	(54)	(45)
Net cash provided by (used in) financing activities	3,853	2,589	(3,527)
Effect of exchange rate changes on cash and cash equivalents and restricted cash	(141)	4	(444)
Net (decrease) increase in cash and cash equivalents and restricted cash	(152)	265	(1,220)
Cash and cash equivalents and restricted cash, beginning of period	17,189	16,924	18,144
Cash and cash equivalents and restricted cash, end of period	\$ 17,037	\$ 17,189	\$ 16,924
Supplemental Non-Cash Investing and Financing Activities			
Acquisitions of property and equipment included in liabilities	\$ 1,410	\$ 2,272	\$ 2,148
Supplemental Disclosures			
Cash paid during the period for interest	\$ 277	\$ 126	\$ 152
Cash paid during the period for income taxes, net of refunds	\$ 1,331	\$ 1,119	\$ 1,203

The accompanying notes are an integral part of these consolidated financial statements.

Revenue Recognition

Revenue by source

The following table disaggregates our revenue by major source (in millions):

	Year Ended December 31,		
	2024	2023	2022
Automotive sales	\$ 72,480	\$ 78,509	\$ 67,210
Automotive regulatory credits	2,763	1,790	1,776
Energy generation and storage sales	9,564	5,515	3,376
Services and other	10,534	8,319	6,091
Total revenues from sales and services	95,341	94,133	78,453
Automotive leasing	1,827	2,120	2,476
Energy generation and storage leasing	522	520	533
Total revenues	\$ 97,690	\$ 96,773	\$ 81,462

Automotive Segment

Automotive Sales

Automotive sales revenue includes revenues related to cash and financing deliveries of new vehicles, and specific other features and services that meet the definition of a performance obligation under ASC 606, including access to our FSD(Supervised) features and their ongoing maintenance, internet connectivity, free Supercharging programs and over-the-air software updates. We recognize revenue on automotive sales upon delivery to the customer, which is when the control of a vehicle transfers. Payments are typically received at the point control transfers or in accordance with payment terms customary to the business, except sales we finance for which payments are collected over the contractual loan term. We also recognize a sales return reserve based on historical experience plus consideration for expected future market values, when we offer resale value guarantees or similar buyback terms. Other features and services such as access to our internet connectivity, unlimited free Supercharging and over-the-air software updates are provisioned upon control transfer of a vehicle and recognized over time on a straight-line basis as we have a stand-ready obligation to deliver such services to the customer. Other limited free Supercharging incentives are recognized based on actual usage or expiration, whichever is earlier. We recognize revenue related to these other features and services over the performance period, which is generally the expected ownership life of the vehicle. Revenue related to FSD (Supervised) features is recognized when functionality is delivered to the customer and their ongoing maintenance is recognized over time. For our obligations related to automotive sales, we estimate standalone selling price by considering costs used to develop and deliver the service, third-party pricing of similar options and other information that may be available.

Any fees that are paid or payable by us to a customer's lender when we arrange the financing are recognized upfront as an offset against automotive sales revenue. Costs to obtain a contract mainly relate to commissions for the sale of vehicles. As our contract costs related to automotive sales are typically fulfilled within one year, the costs to obtain a contract are expensed as incurred. Amounts billed to customers related to shipping and handling are classified as automotive sales revenue, and we have elected to recognize the cost for freight and shipping when control over vehicles, parts or accessories have transferred to the customer as an expense in cost of automotive sales revenue. Our policy is to exclude taxes collected from a customer from the transaction price of automotive contracts.

We offer resale value guarantees to our commercial banking partners in connection with certain vehicle leasing programs. Under these programs, we originate the lease with our end customer and immediately transfer the lease and the underlying vehicle to our commercial banking partner, with the transaction being accounted for as a sale under ASC 606.

We receive upfront payment for the vehicle, do not bear casualty and credit risks during the lease term, and we provide a guarantee capped to a limit if they are unable to sell the vehicle at or above the vehicle's contractual or determined residual value at the end of the lease term. We estimate a guarantee liability in accordance with ASC 460, *Guarantees* and record it within other liabilities on our consolidated balance sheet. On a quarterly basis, we assess the estimated market value of vehicles sold under these programs to determine whether there have been changes to the amount of expected resale value guarantee liabilities. As we accumulate more data related to the resale values of our vehicles or as market conditions change, there may be material changes to their estimated values. The total recorded guarantee liabilities on vehicles sold under these programs were immaterial as of December 31, 2024 and 2023. Our maximum exposure on the guarantees we provide if they are unable to sell the vehicle at or above the vehicle's contractual residual value at the end of the lease term was \$1.45 billion and \$166 million as of December 31, 2024 and 2023, respectively.

Deferred revenue related to the access to our FSD (Supervised) features and their ongoing maintenance, internet connectivity, free Supercharging programs and over-the-air software updates primarily on automotive sales consisted of the following (in millions):

	Year Ended December 31,	
	2024	2023
Deferred revenue — beginning of period	\$ 3,536	\$ 2,913
Additions	1,343	1,201
Net changes in liability for pre-existing contracts, including foreign exchange impact	(92)	17
Revenue recognized	(1,188)	(595)
Deferred revenue — end of period	\$ 3,599	\$ 3,536

Deferred revenue is equivalent to the total transaction price allocated to the performance obligations that are unsatisfied, or partially unsatisfied, as of the balance sheet date. Revenue recognized from the deferred revenue balance as of December 31, 2023 and 2022 was \$872 million and \$469 million for the years ended December 31, 2024 and 2023, respectively. Of the total deferred revenue balance as of December 31, 2024, we expect to recognize \$840 million of revenue in the next 12 months. The remaining balance will be recognized at the time of transfer of control of the product or over the performance period as discussed above in Automotive Sales.

We have financing receivables on our consolidated balance sheets related to loans we provide for financing our automotive deliveries. As of December 31, 2024 and 2023, we had current net financing receivables of \$247 million and \$242 million, respectively, in Accounts receivable, net, and \$821 million and \$1.04 billion, respectively, in Other non-current assets for the long-term portion.

Automotive Regulatory Credits

We earn tradable credits in the operation of our automotive business under various regulations related to ZEVs, greenhouse gas, fuel economy and clean fuel. We sell these credits to other regulated entities who can use the credits to comply with emission standards and other regulatory requirements.

Payments for automotive regulatory credits are typically received at the point control transfers to the customer, or in accordance with payment terms customary to the business. We recognize revenue on the sale of automotive regulatory credits, which have negligible incremental costs associated with them, at the time control of the regulatory credits is transferred to the purchasing party. Deferred revenue related to sales of automotive regulatory credits was immaterial as of December 31, 2024 and 2023. Revenue recognized from the deferred revenue balance as of December 31, 2023 and 2022 was immaterial for the years ended December 31, 2024 and 2023. During the year ended December 31, 2022, we had also recognized \$288 million in revenue due to changes in regulation which entitled us to additional consideration for credits sold previously. We have elected the practical expedient to omit disclosure of the amount of the transaction price allocated to remaining performance obligations for contracts with an original expected contract length of one year or less. As of December 31, 2024, total transaction price allocated to performance obligations that were unsatisfied or partially unsatisfied for contracts with an original expected length of more than one year was \$4.68 billion. Of this amount, we expect to recognize \$863 million in the next 12 months and the rest over the remaining performance obligation period. Additionally, changes in regulations on automotive regulatory credits may significantly impact our remaining performance obligations and revenue to be recognized under these contracts.

We capitalize initial direct costs from the execution of agreements for solar energy systems and PPAs, which include the referral fees and sales commissions, as an element of solar energy systems, net, and subsequently amortize these costs over the term of the related agreements.

Cost of Revenues

Automotive Segment

Automotive Sales

Cost of automotive sales revenue includes direct and indirect materials, labor costs, manufacturing overhead, including depreciation costs of tooling and machinery, shipping and logistic costs, vehicle connectivity costs, FSD (Supervised) ongoing maintenance costs, electricity costs for our free Supercharging programs and reserves for estimated warranty expenses. Cost of automotive sales revenues also includes adjustments to warranty expense and charges to write down the carrying value of our inventory when it exceeds its estimated net realizable value and to provide for obsolete and on-hand inventory in excess of forecasted demand. Additionally, cost of automotive sales revenue benefits from manufacturing credits earned, amounting to \$625 million and \$359 million for the years ended December 31, 2024 and 2023, respectively.

Automotive Leasing

Cost of automotive leasing revenue includes the depreciation of operating lease vehicles, cost of goods sold associated with direct sales-type leases and warranty expense related to leased vehicles.

Services and Other

Costs of services and other revenue includes cost of used vehicles including refurbishment costs, costs associated with providing non-warranty after-sales services, costs associated with our body shops and part sales, costs of paid Supercharging, costs to provide vehicle insurance and costs of retail merchandise sales.

Energy Generation and Storage Segment

Energy Generation and Storage

Cost of energy generation and storage revenue includes direct and indirect material and labor costs, manufacturing overhead, including depreciation costs of tooling and machinery, freight, warranty expense, and cost of servicing. Cost of energy generation and storage revenue also includes charges to write down the carrying value of our inventory when it exceeds its estimated net realizable value and to provide for obsolete and on-hand inventory in excess of forecasted demand. Additionally, cost of energy generation and storage revenue benefits from manufacturing credits earned, amounting to \$756 million and \$115 million for the years ended December 31, 2024 and 2023, respectively. In agreements for solar energy systems and PPAs where we are the lessor, the cost of revenue is primarily comprised of depreciation of the cost of leased solar energy systems, maintenance costs associated with those systems and amortization of any initial direct costs.

Research and Development Costs

Research and development costs are expensed as incurred.

Income Taxes

We are subject to income taxes in the U.S. and in many foreign jurisdictions. Income taxes are computed using the asset and liability method, under which deferred tax assets and liabilities are determined based on the difference between the financial statement and tax bases of assets and liabilities using enacted tax rates in effect for the year in which the differences are expected to affect taxable income. Valuation allowances are established when necessary to reduce deferred tax assets to the amount expected to be realized.

Noncontrolling Interests and Redeemable Noncontrolling Interests

Noncontrolling interests and redeemable noncontrolling interests represent third-party interests in the net assets under certain funding arrangements, or funds, that we have entered into to finance the costs of solar energy systems and vehicles under operating leases. We have determined that the contractual provisions of the funds represent substantive profit-sharing arrangements. We have further determined that the methodology for calculating the noncontrolling interest and redeemable noncontrolling interest balances that reflects the substantive profit-sharing arrangements is a balance sheet approach using the hypothetical liquidation at book value ("HLBV") method. We, therefore, determine the amount of the noncontrolling interests and redeemable noncontrolling interests in the net assets of the funds at each balance sheet date using the HLBV method, which is presented on the consolidated balance sheet as noncontrolling interests in subsidiaries and redeemable noncontrolling interests in subsidiaries. Under the HLBV method, the amounts reported as noncontrolling interests and redeemable noncontrolling interests in the consolidated balance sheet represent the amounts the third parties would hypothetically receive at each balance sheet date under the liquidation provisions of the funds, assuming the net assets of the funds were liquidated at their recorded amounts determined in accordance with GAAP and with tax laws effective at the balance sheet date and distributed to the third parties. The third parties' interests in the results of operations of the funds are determined as the difference in the noncontrolling interest and redeemable noncontrolling interest balances in the consolidated balance sheets between the start and end of each reporting period, after taking into account any capital transactions between the funds and the third parties. However, the redeemable noncontrolling interest balance is at least equal to the redemption amount. The redeemable noncontrolling interest balance is presented as temporary equity in the mezzanine section of the consolidated balance sheet since these third parties have the right to redeem their interests in the funds for cash or other assets.

Net Income per Share of Common Stock Attributable to Common Stockholders

Basic net income per share of common stock attributable to common stockholders is calculated by dividing net income attributable to common stockholders by the weighted-average shares of common stock outstanding for the period. Potentially dilutive shares, which are based on the weighted-average shares of common stock underlying outstanding stock-based awards, warrants and convertible senior notes using the treasury stock method or the if-converted method, as applicable, are included when calculating diluted net income per share of common stock attributable to common stockholders when their effect is dilutive.

The following table presents the reconciliation of net income attributable to common stockholders to net income used in computing basic and diluted net income per share of common stock (in millions):

	Year Ended December 31,		
	2024	2023	2022
Net income attributable to common stockholders	\$ 7,091	\$ 14,997	\$ 12,556
Less: Buy-outs of noncontrolling interests	(39)	(2)	(27)
Net income used in computing basic net income per share of common stock	7,130	14,999	12,583
Less: Dilutive convertible debt	—	—	(1)
Net income used in computing diluted net income per share of common stock	\$ 7,130	\$ 14,999	\$ 12,584

The following table presents the reconciliation of basic to diluted weighted average shares used in computing net income per share of common stock attributable to common stockholders (in millions):

	Year Ended December 31,		
	2024	2023	2022
Weighted average shares used in computing net income per share of common stock, basic	3,197	3,174	3,130
Add:			
Stock-based awards	292	298	310
Convertible senior notes	1	2	3
Warrants	8	11	32
Weighted average shares used in computing net income per share of common stock, diluted	3,498	3,485	3,475

The following table presents the potentially dilutive shares that were excluded from the computation of diluted net income per share of common stock attributable to common stockholders, because their effect was anti-dilutive (in millions):

	Year Ended December 31,		
	2024	2023	2022
Stock-based awards	15	12	4

Cash and Cash Equivalents

All highly liquid investments with an original maturity of three months or less at the date of purchase are considered cash equivalents.

Restricted Cash

We maintain certain cash balances restricted as to withdrawal or use. Our restricted cash is comprised primarily of cash held to service certain payments under various secured debt facilities, cash held as collateral for sales to lease partners with a resale value guarantee and deposits held for our insurance services. We record restricted cash as other assets in the consolidated balance sheets and determine current or non-current classification based on the expected duration of the restriction.

Our total cash and cash equivalents and restricted cash, as presented in the consolidated statements of cash flows, was as follows (in millions):

	December 31, 2024	December 31, 2023	December 31, 2022
Cash and cash equivalents	\$ 16,139	\$ 16,398	\$ 16,253
Restricted cash included in prepaid expenses and other current assets	494	543	294
Restricted cash included in other non-current assets	404	248	377
Total as presented in the consolidated statements of cash flows	\$ 17,037	\$ 17,189	\$ 16,924

Investments

Our investments are all designated as available-for-sale and reported at estimated fair value, with unrealized gains and losses recorded in accumulated other comprehensive loss which is included within stockholders' equity. Available-for-sale marketable securities with maturities greater than three months at the date of purchase are included in short-term investments in our consolidated balance sheets. Interest, dividends, amortization and accretion of purchase premiums and discounts on these investments are included within Interest income in our consolidated statements of operations. Interest income on our short-term investments for the years ended December 31, 2024, 2023 and 2022 was \$763 million, \$388 million and an immaterial amount, respectively.

For periods prior to January 1, 2024, impairment losses were recognized within Restructuring and other in the consolidated statements of operations in the period in which the impairment was identified. Also for periods prior to January 1, 2024, gains were not recorded until realized upon sale(s), at which point they were presented net of any impairment losses for the same digital assets held within Restructuring and other. In determining the gain to be recognized upon sale, we calculate the difference between the sales price and carrying value of the digital assets sold immediately prior to sale.

See Note 3, *Digital Assets, Net*, for further information regarding digital assets.

Solar Energy Systems, Net

We are the lessor of solar energy systems, which are stated at cost less accumulated depreciation. Depreciation of solar energy systems is calculated using the straight-line method over the estimated useful lives of 30 to 35 years. The gross cost of solar energy systems as of December 31, 2024 and 2023 was \$6.79 billion and \$6.87 billion, respectively. Solar energy systems on the consolidated balance sheets are presented net of accumulated depreciation of \$1.86 billion and \$1.64 billion as of December 31, 2024 and 2023, respectively.

Property, Plant and Equipment, Net

Property, plant and equipment, net, including leasehold improvements, are recognized at cost less accumulated depreciation. Depreciation is generally computed using the straight-line method over the estimated useful lives of the respective assets, as follows:

Machinery, equipment, vehicles and office furniture	3 to 15 years
Tooling	4 to 7 years
Building and building improvements	15 to 30 years
Computer equipment and software	3 to 6 years
AI infrastructure	5 to 30 years

Leasehold improvements are depreciated on a straight-line basis over their estimated useful lives. AI infrastructure includes our owned data centers.

Upon the retirement or sale of our property, plant and equipment, the cost and associated accumulated depreciation are removed from the consolidated balance sheet, and the resulting gain or loss is reflected on the consolidated statement of operations. Maintenance and repair expenditures are expensed as incurred while major improvements that increase the functionality, output or expected life of an asset are capitalized and depreciated ratably over the identified useful life.

Long-Lived Assets Including Acquired Intangible Assets

We review our property, plant and equipment, solar energy systems, long-term prepayments and intangible assets for impairment whenever events or changes in circumstances indicate that the carrying amount of an asset (or asset group) may not be recoverable. We measure recoverability by comparing the carrying amount to the future undiscounted cash flows that the asset is expected to generate. If the asset is not recoverable, its carrying amount would be adjusted down to its fair value. For the years ended December 31, 2024, 2023 and 2022, we have recognized no material impairments of our long-lived assets.

Intangible assets with definite lives are amortized on a straight-line basis over their estimated useful lives, which range from seven to thirty years.

Goodwill

We assess goodwill for impairment annually in the fourth quarter, or more frequently if events or changes in circumstances indicate that it might be impaired, by comparing its carrying value to the reporting unit's fair value. For the years ended December 31, 2024, 2023, and 2022, we did not recognize any impairment of goodwill.

Capitalization of Software Costs

We capitalize costs incurred in the development of internal use software, during the application development stage to Property, plant and equipment, net on the consolidated balance sheets. Costs related to preliminary project activities and post-implementation activities are expensed as incurred. Such costs are amortized on a straight-line basis over their estimated useful life of three to five years.

Software development costs incurred in development of software to be sold, leased, or otherwise marketed, incurred subsequent to the establishment of technological feasibility and prior to the general availability of the software are capitalized when they are expected to become significant. Such costs are amortized over the estimated useful life of the applicable software once it is made generally available to our customers.

We evaluate the useful lives of these assets on an annual basis, and we test for impairment whenever events or changes in circumstances occur that could impact the recoverability of these assets. For the years ended December 31, 2024, 2023, and 2022, we have recognized no impairments of capitalized software costs.

Foreign Currency

We determine the functional and reporting currency of each of our international subsidiaries and their operating divisions based on the primary currency in which they operate. In cases where the functional currency is not the U.S. dollar, we recognize a cumulative translation adjustment created by the different rates we apply to current period income or loss and the balance sheet. For each subsidiary, we apply the monthly average functional exchange rate to its monthly income or loss and the month-end functional currency rate to translate the balance sheet.

Foreign currency transaction gains and losses are a result of the effect of exchange rate changes on transactions denominated in currencies other than the functional currency of the respective subsidiary. Transaction gains and losses are recognized in Other income (expense), net, in the consolidated statements of operations. For the years ended December 31, 2024, 2023 and 2022, we recorded a net foreign currency transaction gain of \$57 million, gain of \$122 million and loss of \$89 million, respectively.

Warranties

We provide a manufacturer's warranty on all new and used vehicles and a warranty on the installation and components of the energy generation and storage systems we sell for periods typically between 10 to 25 years. We accrue a warranty reserve for the products sold by us, which includes our best estimate of the projected costs to repair or replace items under warranties and recalls if identified. These estimates are based on actual claims incurred to date and an estimate of the nature, frequency and costs of future claims. These estimates are inherently uncertain and changes to our historical or projected warranty experience may cause material changes to the warranty reserve in the future. The warranty reserve does not include projected warranty costs associated with our vehicles subject to operating lease accounting and our solar energy systems under lease contracts or PPAs, as the costs to repair these warranty claims are expensed as incurred. The portion of the warranty reserve expected to be incurred within the next 12 months is included within Accrued liabilities and other, while the remaining balance is included within Other long-term liabilities on the consolidated balance sheets. For liabilities that we are entitled to receive indemnification from our suppliers, we record receivables for the contractually obligated amounts on the consolidated balance sheets as a component of Prepaid expenses and other current assets for the current portion and as Other non-current assets for the long-term portion. Warranty expense is recorded as a component of Cost of revenues in the consolidated statements of operations. Due to the magnitude of our automotive business, our accrued warranty balance is primarily related to our automotive segment. Accrued warranty activity consisted of the following (in millions):

	Year Ended December 31,		
	2024	2023	2022
Accrued warranty—beginning of period	\$ 5,152	\$ 3,505	\$ 2,101
Warranty costs incurred	(1,453)	(1,225)	(803)
Net changes in liability for pre-existing warranties, including expirations and foreign exchange impact	287	539	522
Provision for warranty	2,730	2,333	1,685
Accrued warranty—end of period	<u>\$ 6,716</u>	<u>\$ 5,152</u>	<u>\$ 3,505</u>

ASU 2023-08

In December 2023, the FASB issued ASU No. 2023-08, Accounting for and Disclosure of Crypto Assets (Subtopic 350-60) (“new crypto assets standard”). The new crypto assets standard requires certain crypto assets to be measured at fair value separately on the balance sheet with changes reported in the statement of operations each reporting period. The new crypto assets standard also enhances the other intangible asset disclosure requirements by requiring the name, cost basis, fair value, and number of units for each significant crypto asset holding. The new crypto assets standard is effective for annual periods beginning after December 15, 2024, including interim periods within those fiscal years. Adoption of the new crypto assets standard requires a cumulative-effect adjustment to the opening balance of retained earnings as of the beginning of the annual reporting period in which an entity adopts the amendments. Early adoption is also permitted, including adoption in an interim period. However, if the new crypto assets standard is early adopted in an interim period, an entity must adopt the new crypto assets standard as of the beginning of the fiscal year that includes the interim period. We adopted the new crypto assets standard on a modified retrospective approach effective January 1, 2024. Refer to Note 3, *Digital Assets, Net* for the inclusion of the new required disclosures. The cumulative effect of the changes made on our January 1, 2024 consolidated balance sheet for the adoption of the new crypto assets standard was as follows (in millions):

	Balances at December 31, 2023	Adjustments from Adoption of the New Crypto Assets Standard	Balances at January 1, 2024
Assets			
Digital assets, net	\$ 184	\$ 303	\$ 487
Deferred tax assets	6,733	(67)	6,666
Stockholders' equity			
Retained earnings	27,882	236	28,118

Note 3 - Digital Assets, Net

During the years ended December 31, 2024 and 2023, we purchased and/or received immaterial amounts of digital assets. During the years ended December 31, 2023 and 2022, we recorded an immaterial amount and \$204 million of impairment losses on digital assets, respectively. The table below summarizes the amounts shown on our consolidated balance sheet as of December 31, 2024 (in millions except units of digital assets).

	December 31, 2024		
	Units	Cost Basis	Fair Value
Digital assets held:			
Bitcoin	11,509	\$ 386	\$ 1,074
Other		1	2
Total		387	1,076

The following table provides details of the activities related to our digital assets for the year ended December 31, 2024 (in millions):

Beginning balance at fair value	\$ 487
Unrealized gains, net	589
Ending balance	\$ 1,076

The following tables present summarized unaudited quarterly financial statement data for those impacted line items based on the Company’s adoption of the new crypto assets standard as described in Note 2, *Summary of Significant Accounting Policies* (in millions, except per share data):

Consolidated Balance Sheets (unaudited):	As of:		
	March 31, 2024	June 30, 2024	September 30, 2024
Digital assets, net			
Before adoption	\$ 184	\$ 184	\$ 184
Adjustments	638	538	545
As adjusted	\$ 822	\$ 722	\$ 729
Deferred tax assets			
Before adoption	\$ 6,769	\$ 6,692	\$ 6,486
Adjustments	(141)	(119)	(120)
As adjusted	\$ 6,628	\$ 6,573	\$ 6,366
Condensed Consolidated Statements of Operations (unaudited):	Three Months Ended		
	March 31, 2024	June 30, 2024	September 30, 2024
Other income (expense), net			
Before adoption	\$ 108	\$ 20	\$ (270)
Adjustments	335	(100)	7
As adjusted	\$ 443	\$ (80)	\$ (263)
Provision for (benefit from) income taxes			
Before adoption	\$ 409	\$ 393	\$ 601
Adjustments	74	(22)	1
As adjusted	\$ 483	\$ 371	\$ 602
Net income attributable to common stockholders			
Before adoption	\$ 1,129	\$ 1,478	\$ 2,167
Adjustments	261	(78)	6
As adjusted	\$ 1,390	\$ 1,400	\$ 2,173
Net income per share of common stock attributable to common stockholders			
Basic			
Before adoption	\$ 0.37	\$ 0.46	\$ 0.68
Adjustments	0.08	(0.02)	—
As adjusted	\$ 0.45	\$ 0.44	\$ 0.68
Diluted			
Before adoption	\$ 0.34	\$ 0.42	\$ 0.62
Adjustments	0.07	(0.02)	—
As adjusted	\$ 0.41	\$ 0.40	\$ 0.62

The unaudited impact of adoption for the three months ended December 31, 2024 was to increase Digital assets, net by \$347 million, with a corresponding increase in Other income (expense), net and decrease our Deferred tax assets by \$77 million with a corresponding increase in our Provision for (benefit from) income taxes, thus contributing \$270 million to Net income attributable to common stockholders. These amounts contributed \$0.08 to both basic and diluted net income attributable to common stockholders per share of common stock for the three months ended December 31, 2024.

There were no impairment losses recorded for any period during the year ended December 31, 2024. As a result, the unaudited interim balances of Digital assets, net did not change throughout the year ended December 31, 2024 prior to the adoption of the new crypto assets standard, which was adopted as of January 1, 2024.

Note 4 - Goodwill and Intangible Assets

Goodwill decreased \$9 million within the automotive segment from \$253 million as of December 31, 2023 to \$244 million as of December 31, 2024. There were no accumulated impairment losses as of December 31, 2024 and 2023.

The net carrying value of our intangible assets decreased from \$178 million as of December 31, 2023 to \$150 million as of December 31, 2024 mainly from amortization.

Note 5 - Fair Value of Financial Instruments

ASC 820 states that fair value is an exit price, representing the amount that would be received to sell an asset or paid to transfer a liability in an orderly transaction between market participants. As such, fair value is a market-based measurement that should be determined based on assumptions that market participants would use in pricing an asset or a liability. The three-tiered fair value hierarchy, which prioritizes which inputs should be used in measuring fair value, is comprised of: (Level I) observable inputs such as quoted prices in active markets; (Level II) inputs other than quoted prices in active markets that are observable either directly or indirectly and (Level III) unobservable inputs for which there is little or no market data. The fair value hierarchy requires the use of observable market data when available in determining fair value. Our assets and liabilities that were measured at fair value on a recurring basis were as follows (in millions):

	December 31, 2024				December 31, 2023			
	Fair Value	Level I	Level II	Level III	Fair Value	Level I	Level II	Level III
Certificates of deposit and time deposits	\$ 12,767	\$ —	\$ 12,767	\$ —	\$ 6,996	\$ —	\$ 6,996	\$ —
Commercial paper	3,919	—	3,919	—	470	—	470	—
U.S. government securities	3,620	—	3,620	—	5,136	—	5,136	—
Corporate debt securities	118	—	118	—	480	—	480	—
Money market funds	1,753	1,753	—	—	109	109	—	—
Digital assets	1,076	1,076	—	—	Not applicable			
Total	\$ 23,253	\$ 2,829	\$ 20,424	\$ —	\$ 13,191	\$ 109	\$ 13,082	\$ —

All of our money market funds and digital assets are classified within Level I of the fair value hierarchy because they were valued using quoted prices in active markets. Our U.S. government securities, certificates of deposit, commercial paper, time deposits and corporate debt securities are classified within Level II of the fair value hierarchy and the market approach was used to determine fair value of these investments.

Our cash, cash equivalents and investments classified by security type as of December 31, 2024 and 2023 consisted of the following (in millions):

	December 31, 2024					
	Adjusted Cost	Gross Unrealized Gains	Gross Unrealized Losses	Fair Value	Cash and Cash Equivalents	Short-Term Investments
Cash	\$ 14,386	\$ —	\$ —	\$ 14,386	\$ 14,386	\$ —
Certificates of deposit and time deposits	12,767	—	—	12,767	—	12,767
Commercial paper	3,908	11	—	3,919	—	3,919
U.S. government securities	3,618	3	(1)	3,620	—	3,620
Corporate debt securities	117	1	—	118	—	118
Money market funds	1,753	—	—	1,753	1,753	—
Total cash, cash equivalents and short-term investments	\$ 36,549	\$ 15	\$ (1)	\$ 36,563	\$ 16,139	\$ 20,424

December 31, 2023						
	Adjusted Cost	Gross Unrealized Gains	Gross Unrealized Losses	Fair Value	Cash and Cash Equivalents	Short-Term Investments
Cash	\$ 15,903	\$ —	\$ —	\$ 15,903	\$ 15,903	\$ —
Certificates of deposit and time deposits	6,995	1	—	6,996	—	6,996
U.S. government securities	5,136	1	(1)	5,136	277	4,859
Corporate debt securities	485	1	(6)	480	—	480
Commercial paper	470	—	—	470	109	361
Money market funds	109	—	—	109	109	—
Total cash, cash equivalents and short-term investments	\$ 29,098	\$ 3	\$ (7)	\$ 29,094	\$ 16,398	\$ 12,696

As of December 31, 2024, the vast majority of our short-term investments had contractual maturity dates within one year.

Disclosure of Fair Values

Our financial instruments that are not re-measured at fair value include accounts receivable, financing receivables, other receivables, accounts payable, accrued liabilities, customer deposits and debt. The carrying values of these financial instruments materially approximate their fair values.

Note 6 - Inventory

Our inventory consisted of the following (in millions):

	December 31, 2024	December 31, 2023
Raw materials	\$ 5,242	\$ 5,390
Work in process	1,532	2,016
Finished goods (1)	3,940	5,049
Service parts	1,303	1,171
Total	\$ 12,017	\$ 13,626

(1) Finished goods inventory includes products-in-transit to fulfill customer orders, new vehicles, used vehicles and energy products available for sale.

We write-down inventory for any excess or obsolete inventory or when we believe that the net realizable value of inventory is less than the carrying value. During the years ended December 31, 2024, 2023 and 2022, we recorded write-downs of \$190 million, \$233 million and \$144 million, respectively, in Cost of revenues in the consolidated statements of operations.

Note 7 - Property, Plant and Equipment, Net

Our property, plant and equipment, net, consisted of the following (in millions):

	December 31, 2024	December 31, 2023
Machinery, equipment, vehicles and office furniture	\$ 18,339	\$ 16,309
Land and buildings	10,677	9,498
AI infrastructure	5,152	1,510
Tooling	3,883	3,129
Leasehold improvements	3,688	3,136
Computer equipment, hardware and software	2,902	2,409
Construction in progress	6,783	5,791
	51,424	41,782
Less: Accumulated depreciation	(15,588)	(12,057)
Total	\$ 35,836	\$ 29,725

Construction in progress is primarily comprised of ongoing construction and expansion of our facilities, equipment and tooling related to the manufacturing of our products as well as AI-related assets which have not yet been placed in service. Completed assets are transferred to their respective asset classes and depreciation begins when an asset is ready for its intended use.

Depreciation expense during the years ended December 31, 2024, 2023 and 2022 was \$4.12 billion, \$3.33 billion and \$2.42 billion, respectively.

Note 8 - Accrued Liabilities and Other

Our accrued liabilities and other current liabilities consisted of the following (in millions):

	December 31, 2024	December 31, 2023
Accrued purchases (1)	\$ 2,253	\$ 2,721
Accrued warranty reserve, current portion	1,917	1,546
Payroll and related costs	1,532	1,325
Taxes payable (2)	1,367	1,204
Customer deposits	993	876
Operating lease liabilities, current portion	807	672
Sales return reserve, current portion	305	219
Other current liabilities	1,549	517
Total	\$ 10,723	\$ 9,080

- (1) Accrued purchases primarily reflects receipts of goods and services for which we had not yet been invoiced. As we are invoiced for these goods and services, this balance will reduce and accounts payable will increase.
- (2) Taxes payable primarily includes value added tax, income tax, sales tax, property tax and use tax payables.

Note 9 - Other Long-Term Liabilities

Our other long-term liabilities consisted of the following (in millions):

	December 31, 2024	December 31, 2023
Operating lease liabilities	\$ 4,603	\$ 3,671
Accrued warranty reserve	4,799	3,606
Other non-current liabilities	1,093	876
Total other long-term liabilities	<u>\$ 10,495</u>	<u>\$ 8,153</u>

Note 10 - Debt

The following is a summary of our debt and finance leases as of December 31, 2024 (in millions):

	Net Carrying Value		Unpaid Principal Balance	Unused Committed Amount (1)	Contractual Interest Rates	Contractual Maturity Date
	Current	Long-Term				
Recourse debt:						
RCF Credit Agreement	—	—	—	5,000	Not applicable	January 2028
Other	4	3	7	—	4.70-5.75%	March 2025-January 2031
Total recourse debt	4	3	7	5,000		
Non-recourse debt:						
Automotive Asset-backed Notes	2,255	2,059	4,329	—	3.45-6.57%	September 2025-June 2035
China Working Capital Facility	—	2,740	2,740	—	1.92 %	April 2025 (2)
Energy Asset-backed Notes	54	434	493	—	4.80-6.25%	December 2025-June 2050
Cash Equity Debt	30	299	338	—	5.25-5.81%	July 2033-January 2035
Total non-recourse debt	2,339	5,532	7,900	—		
Total debt	2,343	5,535	\$ 7,907	\$ 5,000		
Finance leases	113	222				
Total debt and finance leases	\$ 2,456	\$ 5,757				

The following is a summary of our debt and finance leases as of December 31, 2023 (in millions):

	Net Carrying Value		Unpaid Principal Balance	Unused Committed Amount (1)	Contractual Interest Rates	Contractual Maturity Date
	Current	Long-Term				
Recourse debt:						
2024 Notes	\$ 37	\$ —	\$ 37	\$ —	2.00 %	May 2024
RCF Credit Agreement	—	—	—	5,000	Not applicable	January 2028
Other	—	7	7	28	4.70-5.75%	March 2025-January 2031
Total recourse debt	37	7	44	5,028		
Non-recourse debt:						
Automotive Asset-backed Notes	1,906	2,337	4,259	—	0.60-6.57%	July 2024-May 2031
Cash Equity Debt	28	330	367	—	5.25-5.81%	July 2033-January 2035
Energy Asset-backed Notes	4	8	13	—	4.80 %	December 2026
Total non-recourse debt	1,938	2,675	4,639	—		
Total debt	1,975	2,682	\$ 4,683	\$ 5,028		
Finance leases	398	175				
Total debt and finance leases	\$ 2,373	\$ 2,857				

- There are no restrictions on draw-down or use for general corporate purposes with respect to any available committed funds under our RCF Credit Agreement, except certain specified conditions prior to draw-down. Refer to the section below for the terms of the facility.
- The contractual maturity date of the China Working Capital Facility is April 2025, renewable until March 2026 at our discretion. As we have the intent and ability to refinance the loan on a long-term basis, we recorded it in Debt and finance leases, net of current portion in the consolidated balance sheets.

China Working Capital Facility

In April 2024, one of our subsidiaries entered into a loan agreement (the “China Working Capital Facility”) with lenders in China for an unsecured revolving facility of up to RMB 20.00 billion to be used for certain production expenditures as well as repayment of certain finance facilities. Borrowed funds bear interest at a rate equal to the Loan Prime Rate published by the People’s Bank of China minus 1.18%. The China Working Capital Facility is non-recourse to our assets.

Pledged Assets

As of December 31, 2024 and 2023, we had pledged or restricted \$5.16 billion and \$4.64 billion of our assets (consisting principally of operating lease vehicles, financing receivables, restricted cash, and equity interests in certain SPEs) as collateral for our outstanding debt.

Schedule of Principal Maturities of Debt

The future scheduled principal maturities of debt as of December 31, 2024 were as follows (in millions):

	Recourse debt	Non-recourse debt	Total
2025	\$ 4	\$ 2,349	\$ 2,353
2026	—	4,116	4,116
2027	—	699	699
2028	—	243	243
2029	—	95	95
Thereafter	3	398	401
Total	\$ 7	\$ 7,900	\$ 7,907

Note 11 - Leases

We have entered into various operating and finance lease agreements for certain of our offices, manufacturing and warehouse facilities, retail and service locations, data centers, equipment, vehicles, and solar energy systems, worldwide. We determine if an arrangement is a lease, or contains a lease, at inception and record the lease in our financial statements upon lease commencement, which is the date when the underlying asset is made available for use by the lessor.

We have lease agreements with lease and non-lease components, and have elected to utilize the practical expedient to account for lease and non-lease components together as a single combined lease component, from both a lessee and lessor perspective with the exception of direct sales-type leases and production equipment classes embedded in supply agreements. From a lessor perspective, the timing and pattern of transfer are the same for the non-lease components and associated lease component and, the lease component, if accounted for separately, would be classified as an operating lease.

We have elected not to present short-term leases on the consolidated balance sheet as these leases have a lease term of 12 months or less at lease inception and do not contain purchase options or renewal terms that we are reasonably certain to exercise. All other lease assets and lease liabilities are recognized based on the present value of lease payments over the lease term at commencement date. Because most of our leases do not provide an implicit rate of return, we used our incremental borrowing rate based on the information available at lease commencement date in determining the present value of lease payments.

Our leases, where we are the lessee, often include options to extend the lease term for up to 10 years. Some of our leases also include options to terminate the lease prior to the end of the agreed upon lease term. For purposes of calculating lease liabilities, lease terms include options to extend or terminate the lease when it is reasonably certain that we will exercise such options.

Lease expense for operating leases is recognized on a straight-line basis over the lease term as cost of revenues or operating expenses depending on the nature of the leased asset. Certain operating leases provide for annual increases to lease payments based on an index or rate. We calculate the present value of future lease payments based on the index or rate at the lease commencement date for new leases. Differences between the calculated lease payment and actual payment are expensed as incurred. Amortization of finance lease assets is recognized over the lease term as cost of revenues or operating expenses depending on the nature of the leased asset. Interest expense on finance lease liabilities is recognized over the lease term within Interest expense in the consolidated statements of operations.

The balances for the operating and finance leases where we are the lessee are presented as follows (in millions) within our consolidated balance sheets:

	December 31, 2024	December 31, 2023
Operating leases:		
Operating lease right-of-use assets	\$ 5,160	\$ 4,180
Accrued liabilities and other	\$ 807	\$ 672
Other long-term liabilities	4,603	3,671
Total operating lease liabilities	<u>\$ 5,410</u>	<u>\$ 4,343</u>
Finance leases:		
Solar energy systems, net	\$ 21	\$ 23
Property, plant and equipment, net	350	601
Total finance lease assets	<u>\$ 371</u>	<u>\$ 624</u>
Current portion of long-term debt and finance leases	\$ 113	\$ 398
Long-term debt and finance leases, net of current portion	222	175
Total finance lease liabilities	<u>\$ 335</u>	<u>\$ 573</u>

The components of lease expense are as follows (in millions) within our consolidated statements of operations:

	Year Ended December 31,		
	2024	2023	2022
Operating lease expense:			
Operating lease expense (1)	\$ 1,500	\$ 1,153	\$ 798
Finance lease expense:			
Amortization of leased assets	\$ 409	\$ 506	\$ 493
Interest on lease liabilities	21	45	72
Total finance lease expense	<u>\$ 430</u>	<u>\$ 551</u>	<u>\$ 565</u>
Total lease expense	<u>\$ 1,930</u>	<u>\$ 1,704</u>	<u>\$ 1,363</u>

(1) Includes short-term leases and variable lease costs, which are immaterial.

Other information related to leases where we are the lessee is as follows:

	December 31, 2024	December 31, 2023
Weighted-average remaining lease term:		
Operating leases	7.9 years	7.4 years
Finance leases	4.4 years	2.3 years
Weighted-average discount rate:		
Operating leases	5.3 %	5.6 %
Finance leases	4.7 %	5.5 %

Supplemental cash flow information related to leases where we are the lessee is as follows (in millions):

	Year Ended December 31,		
	2024	2023	2022
Cash paid for amounts included in the measurement of lease liabilities:			
Operating cash outflows from operating leases	\$ 1,450	\$ 1,084	\$ 754
Operating cash outflows from finance leases (interest payments)	\$ 21	\$ 47	\$ 75
Leased assets obtained in exchange for finance lease liabilities	\$ 144	\$ 10	\$ 58
Leased assets obtained in exchange for operating lease liabilities	\$ 1,808	\$ 2,170	\$ 1,059

As of December 31, 2024, the maturities of our operating and finance lease liabilities (excluding short-term leases) are as follows (in millions):

	Operating Leases	Finance Leases
2025	\$ 1,066	\$ 122
2026	958	84
2027	843	69
2028	742	27
2029	642	22
Thereafter	2,428	45
Total minimum lease payments	6,679	369
Less: Interest	1,269	34
Present value of lease obligations	5,410	335
Less: Current portion	807	113
Long-term portion of lease obligations	\$ 4,603	\$ 222

As of December 31, 2024, we have excluded from the table above additional operating leases that have not yet commenced with aggregate rent payments of \$1.10 billion. These operating leases will commence between fiscal year 2025 and 2026 with lease terms of 2 years to 20 years.

Operating Lease and Sales-type Lease Receivables

We are the lessor of certain vehicle and solar energy system arrangements as described in Note 2, *Summary of Significant Accounting Policies*. As of December 31, 2024, maturities of our operating lease and sales-type lease receivables from customers for each of the next five years and thereafter were as follows (in millions):

	Operating Leases	Sales-type Leases
2025	\$ 1,265	\$ 177
2026	777	185
2027	454	93
2028	240	27
2029	200	2
Thereafter	1,290	—
Gross lease receivables	<u>\$ 4,226</u>	<u>\$ 484</u>

The above table does not include vehicle sales to customers or leasing partners with a resale value guarantee as the cash payments were received upfront. For our solar PPA arrangements, customers are charged solely based on actual power produced by the installed solar energy system at a predefined rate per kilowatt-hour of power produced. The future payments from such arrangements are not included in the above table as they are a function of the power generated by the related solar energy systems in the future.

Net Investment in Sales-type Leases

Net investment in sales-type leases, which is the sum of the present value of the future contractual lease payments, is presented on the consolidated balance sheets as a component of Prepaid expenses and other current assets for the current portion and as Other non-current assets for the long-term portion. Lease receivables relating to sales-type leases are presented on the consolidated balance sheets as follows (in millions):

	December 31, 2024	December 31, 2023
Gross lease receivables	\$ 484	\$ 780
Unearned interest income	(38)	(78)
Allowance for expected credit losses	(6)	(6)
Net investment in sales-type leases	<u>\$ 440</u>	<u>\$ 696</u>
Reported as:		
Prepaid expenses and other current assets	\$ 152	\$ 189
Other non-current assets	288	507
Net investment in sales-type leases	<u>\$ 440</u>	<u>\$ 696</u>

Note 12 - Equity Incentive Plans

In June 2019, we adopted the 2019 Equity Incentive Plan (the “2019 Plan”). The 2019 Plan provides for the grant of stock options, restricted stock, RSUs, stock appreciation rights, performance units and performance shares to our employees, directors and consultants. Stock options granted under the 2019 Plan may be either incentive stock options or nonstatutory stock options. Incentive stock options may only be granted to our employees. Nonstatutory stock options may be granted to our employees, directors and consultants. Generally, our stock options and RSUs vest over four years and our stock options are exercisable over a maximum period of 10 years from their grant dates. Vesting typically terminates when the employment or consulting relationship ends.

As of December 31, 2024, 113.0 million shares were reserved and available for issuance under the 2019 Plan.

The following table summarizes our stock option and RSU activity for the year ended December 31, 2024:

	Stock Options				RSUs	
	Number of Options (in thousands)	Weighted-Average Exercise Price	Weighted-Average Remaining Contractual Life (years)	Aggregate Intrinsic Value (in billions)	Number of RSUs (in thousands)	Weighted-Average Grant Date Fair Value
Beginning of period	344,021	\$ 35.11			19,088	\$ 225.01
Granted	14,979	\$ 235.22			12,045	\$ 223.98
Exercised or released	(10,469)	\$ 82.04			(7,515)	\$ 206.02
Cancelled	(3,996)	\$ 205.61			(4,957)	\$ 221.15
End of period	344,535	\$ 40.41	3.51	\$ 125.22	18,661	\$ 233.02
Vested and expected to vest, December 31, 2024	344,215	\$ 40.23	3.50	\$ 125.16	18,409	\$ 232.99
Exercisable and vested, December 31, 2024	323,847	\$ 27.75	3.14	\$ 121.79		

The weighted-average grant date fair value per share of RSUs granted in the years ended December 31, 2024, 2023 and 2022 was \$223.98, \$228.33 and \$239.85, respectively. The aggregate release date fair value of RSUs in the years ended December 31, 2024, 2023 and 2022 was \$1.75 billion, \$2.50 billion and \$4.32 billion, respectively.

The aggregate intrinsic value of options exercised during the years ended December 31, 2024, 2023, and 2022 was \$1.77 billion, \$1.33 billion and \$1.90 billion, respectively.

ESPP

Our employees are eligible to purchase our common stock through payroll deductions of up to 15% of their eligible compensation, subject to any plan limitations. The purchase price would be 85% of the lower of the fair market value on the first and last trading days of each six-month offering period. During the years ended December 31, 2024, 2023 and 2022, under the ESPP we issued 2.2 million, 2.1 million and 1.4 million shares, respectively. As of December 31, 2024, there were 95.5 million shares available for issuance under the ESPP.

Fair Value Assumptions

We use the fair value method in recognizing stock-based compensation expense. Under the fair value method, we estimate the fair value of each stock option award with service or service and performance conditions and the ESPP on the grant date generally using the Black-Scholes option pricing model. The weighted-average assumptions used in the Black-Scholes model for stock options are as follows:

	Year Ended December 31,		
	2024	2023	2022
Risk-free interest rate	3.92 %	3.90 %	3.11 %
Expected term (in years)	4.3	4.5	4.1
Expected volatility	59 %	63 %	63 %
Dividend yield	0.0 %	0.0 %	0.0 %
Grant date fair value per share	\$ 114.29	\$ 121.62	\$ 114.51

The fair value of RSUs with service or service and performance conditions is measured on the grant date based on the closing fair market value of our common stock. The risk-free interest rate is based on the U.S. Treasury yield for zero-coupon U.S. Treasury notes with maturities approximating each grant's expected life. We use our historical data in estimating the expected term of our employee grants. The expected volatility is based on the average of the implied volatility of publicly traded options for our common stock and the historical volatility of our common stock.

Other Performance-Based Grants

From time to time, the Compensation Committee of our Board of Directors grants certain employees performance-based RSUs and stock options.

As of December 31, 2024, we had unrecognized stock-based compensation expense of \$490 million under these grants to purchase or receive an aggregate 5.0 million shares of our common stock. For awards probable of achievement, we estimate the unrecognized stock-based compensation expense of \$421 million will be recognized over a weighted-average period of 4.0 years.

For the years ended December 31, 2024, 2023 and 2022, stock-based compensation expense related to these grants, net of forfeitures, were not material.

Summary Stock-Based Compensation Information

The following table summarizes our stock-based compensation expense by line item in the consolidated statements of operations (in millions):

	Year Ended December 31,		
	2024	2023	2022
Cost of revenues	\$ 776	\$ 741	\$ 594
Research and development	832	689	536
Selling, general and administrative	389	382	430
Restructuring and other	\$ 2	\$ —	\$ —
Total	\$ 1,999	\$ 1,812	\$ 1,560

Our income tax benefits recognized from stock-based compensation arrangements were immaterial while we were under full valuation allowances on our U.S. deferred tax assets during the year ended December 31, 2022. With the release of the valuation allowance associated with our federal and certain state deferred tax assets in 2023, income tax benefits recognized from stock-based compensation expense during the years ended December 31, 2024 and 2023 were \$371 million and \$326 million, respectively. During the years ended December 31, 2024, 2023 and 2022, stock-based compensation expense capitalized to our consolidated balance sheets was \$198 million, \$199 million and \$245 million, respectively. As of December 31, 2024, we had \$5.79 billion of total unrecognized stock-based compensation expense related to non-performance awards, which will be recognized over a weighted-average period of 3.0 years.

Note 13 - Income Taxes

Our income before provision for (benefit from) income taxes for the years ended December 31, 2024, 2023 and 2022 was as follows (in millions):

	Year Ended December 31,		
	2024	2023	2022
Domestic	\$ 2,292	\$ 3,196	\$ 5,524
Noncontrolling interest and redeemable noncontrolling interest	62	(23)	31
Foreign	6,636	6,800	8,164
Income before income taxes	\$ 8,990	\$ 9,973	\$ 13,719

A provision for (benefit from) income taxes of \$1.84 billion, \$(5.00) billion and \$1.13 billion has been recognized for the years ended December 31, 2024, 2023 and 2022, respectively. The components of the provision for (benefit from) income taxes for the years ended December 31, 2024, 2023 and 2022 consisted of the following (in millions):

	Year Ended December 31,		
	2024	2023	2022
Current:			
Federal	\$ —	\$ 48	\$ —
State	45	57	62
Foreign	1,315	1,243	1,266
Total current	1,360	1,348	1,328
Deferred:			
Federal	831	(5,246)	26
State	(49)	(653)	1
Foreign	(305)	(450)	(223)
Total deferred	477	(6,349)	(196)
Total provision for (benefit from) income taxes	\$ 1,837	\$ (5,001)	\$ 1,132

The reconciliation of taxes at the federal statutory rate to our provision for (benefit from) income taxes for the years ended December 31, 2024, 2023 and 2022 was as follows (in millions):

	Year Ended December 31,		
	2024	2023	2022
Tax at statutory federal rate	\$ 1,887	\$ 2,094	\$ 2,881
State tax, net of federal benefit	8	(372)	51
Excess tax benefits related to stock-based compensation	(267)	(288)	(745)
Nontaxable manufacturing credit	(291)	(101)	—
Foreign income rate differential	(545)	(816)	(923)
U.S. tax credits	(317)	(593)	(276)
GILTI and Subpart F inclusion	882	670	1,279
Unrecognized tax benefits	144	183	252
Change in valuation allowance	163	(5,962)	(1,532)
Other	173	184	145
Provision for (benefit from) income taxes	\$ 1,837	\$ (5,001)	\$ 1,132

Deferred tax assets (liabilities) as of December 31, 2024 and 2023 consisted of the following (in millions):

	December 31, 2024	December 31, 2023
Deferred tax assets:		
Net operating loss carry-forwards	\$ 1,295	\$ 2,826
Research and development credits	1,735	1,358
Other tax credits and attributes	1,325	827
Deferred revenue	1,101	1,035
Inventory and warranty reserves	1,769	1,258
Operating lease right-of-use liabilities	1,186	930
Capitalized research and development costs	2,448	1,344
Deferred GILTI tax assets	691	760
Other	412	436
Total deferred tax assets	11,962	10,774
Valuation allowance	(1,224)	(892)
Deferred tax assets, net of valuation allowance	10,738	9,882
Deferred tax liabilities:		
Depreciation and amortization	(2,658)	(2,122)
Operating lease right-of-use assets	(1,097)	(859)
Other	(561)	(249)
Total deferred tax liabilities	(4,316)	(3,230)
Deferred tax assets (liabilities), net of valuation allowance	\$ 6,422	\$ 6,652

As of December 31, 2024, we maintained valuation allowances of \$1.22 billion for deferred tax assets that are not more likely than not to be realized, which primarily included our California deferred tax assets, U.S. foreign tax credits and certain foreign operating losses. The valuation allowance on our net deferred tax assets increased by \$332 million during the year ended December 31, 2024, and decreased by \$6.46 billion and \$1.73 billion during the years ended December 31, 2023 and 2022, respectively. The valuation allowance increase during the year ended December 31, 2024 was primarily due to the changes of our California deferred tax assets, U.S. foreign tax credits and certain foreign operating losses. The change in valuation allowance during the year ended December 31, 2023 was primarily due to the release of our valuation allowance with respect to our U.S. federal and certain state deferred tax assets. In the fourth quarter of 2023, based on the relevant weight of positive and negative evidence, including the amount of our taxable income in recent years which was objective and verifiable, and consideration of our expected future taxable earnings, we concluded that it is more likely than not that most of our U.S. federal and certain state deferred tax assets are realizable and released the valuation allowance on these deferred tax assets. The valuation allowance change during the years ended December 31, 2022 was primarily due to changes in our U.S. deferred tax assets and liabilities. Our deferred tax assets without a valuation allowance are more likely than not to be realized given the expectation of future earnings in the respective jurisdictions.

As of December 31, 2024, we had \$4.34 billion of federal and \$8.59 billion of state net operating loss carry-forwards available to offset future taxable income, an immaterial amount of which, if not utilized, will begin to expire in 2026 for federal and 2025 for state purposes. Federal and state laws can impose substantial restrictions on the utilization of net operating loss and tax credit carry-forwards in the event of an "ownership change," as defined in Section 382 of the Internal Revenue Code. We have determined that no significant limitation would be placed on the utilization of our net operating loss and tax credit carry-forwards due to prior ownership changes or expirations.

As of December 31, 2024, we had federal research and development tax credits of \$1.48 billion, federal renewable energy tax credits of \$1.01 billion, and state research and development tax credits of \$1.06 billion. Most of our state research and development tax credits were in the state of California. If not utilized, some of the federal tax credits may expire in various amounts beginning in 2035. However, California research and development tax credits can be carried forward indefinitely.

As of December 31, 2024, we intend to indefinitely reinvest our foreign earnings and cash unless such repatriation results in no or minimal tax costs. We have recorded the taxes associated with the foreign earnings we intend to repatriate in the future. For the earnings we intend to indefinitely reinvest, no deferred tax liabilities for foreign withholding or other taxes have been recorded. The estimated amount of such unrecognized withholding tax liability associated with the indefinitely reinvested earnings is approximately \$309 million.

Uncertain Tax Positions

The changes to our gross unrecognized tax benefits were as follows (in millions):

December 31, 2021	\$	531
Increases in balances related to prior year tax positions		136
Decreases in balances related to prior year tax positions		(12)
Increases in balances related to current year tax positions		222
Decreases in balances related to expiration of the statute of limitations		(7)
December 31, 2022		870
Increases in balances related to prior year tax positions		59
Decreases in balances related to settlement with tax authorities		(6)
Increases in balances related to current year tax positions		255
Decreases in balances related to expiration of the statute of limitations		(4)
December 31, 2023		1,174
Increases in balances related to prior year tax positions		51
Decreases in balances related to prior year tax positions		(27)
Increases in balances related to current year tax positions		227
Decreases in balances related to settlement with tax authorities		(4)
Decreases in balances related to expiration of the statute of limitations		(4)
December 31, 2024	\$	1,417

The aggregate carrying values of the VIEs' assets and liabilities, after elimination of any intercompany transactions and balances, in the consolidated balance sheets were as follows (in millions):

	December 31, 2024	December 31, 2023
Assets		
Current assets		
Cash and cash equivalents	\$ 49	\$ 66
Accounts receivable, net	18	13
Prepaid expenses and other current assets	276	361
Total current assets	343	440
Operating lease vehicles, net	392	—
Solar energy systems, net	2,310	3,278
Other non-current assets	183	369
Total assets	\$ 3,228	\$ 4,087
Liabilities		
Current liabilities		
Accrued liabilities and other	\$ 32	\$ 67
Deferred revenue	6	6
Current portion of debt and finance leases	2,114	1,564
Total current liabilities	2,152	1,637
Deferred revenue, net of current portion	71	99
Debt and finance leases, net of current portion	1,834	2,041
Total liabilities	\$ 4,057	\$ 3,777

Note 16 - Related Party Transactions

Tesla periodically does business with certain entities with which its CEO and directors are affiliated, such as x.AI, SpaceX, The Boring Company, X Corp. and Redwood Materials, in accordance with our Related Person Transactions Policy. Such transactions have not had to date, and are not currently expected to have, a material impact on our consolidated financial statements.

Note 17 - Segment Reporting and Information about Geographic Areas

Our Chief Executive Officer, as the CODM, organizes our company, manages resource allocations and measures performance among two operating and reportable segments: (i) automotive and (ii) energy generation and storage. The automotive segment includes the design, development, manufacturing, sales and leasing of electric vehicles as well as sales of automotive regulatory credits. Additionally, the automotive segment is also comprised of services and other, which includes sales of used vehicles, non-warranty maintenance services and collision, part sales, paid Supercharging, insurance services revenue and retail merchandise sales. The energy generation and storage segment includes the design, manufacture, installation, sales and leasing of solar energy generation and energy storage products and related services and sales of solar energy systems incentives.

Our CODM does not evaluate operating segments using asset or liability information. The CODM uses gross profit to allocate operating and capital resources and assesses performance of each segment by comparing actual gross profit results to historical results and previously forecasted financial information. The following table presents revenues, cost of revenues and gross profit by reportable segment (in millions):

	Year Ended December 31,		
	2024	2023	2022
Automotive segment			
Revenues	\$ 87,604	\$ 90,738	\$ 77,553
Cost of revenues (1)	\$ 72,794	\$ 74,219	\$ 56,988
Gross profit	\$ 14,810	\$ 16,519	\$ 20,565
Energy generation and storage segment			
Revenues	\$ 10,086	\$ 6,035	\$ 3,909
Cost of revenues (2)	\$ 7,446	\$ 4,894	\$ 3,621
Gross profit	\$ 2,640	\$ 1,141	\$ 288

- (1) Depreciation and amortization included in Cost of revenues for the automotive segment for the years ended December 31, 2024, 2023 and 2022 was \$3.68 billion, \$3.45 billion and \$2.81 billion, respectively.
- (2) Depreciation and amortization included in Cost of revenues for the energy generation and storage segment for the years ended December 31, 2024, 2023 and 2022 was \$377 million, \$343 million and \$331 million, respectively.

The following table presents revenues by geographic area based on the sales location of our products (in millions):

	Year Ended December 31,		
	2024	2023	2022
United States	\$ 47,725	\$ 45,235	\$ 40,553
China	20,944	21,745	18,145
Other international	29,021	29,793	22,764
Total	\$ 97,690	\$ 96,773	\$ 81,462

The following table presents long-lived assets by geographic area (in millions):

	December 31, 2024	December 31, 2023
United States	\$ 32,461	\$ 26,629
Germany	4,175	4,258
Other international	4,124	4,067
Total	\$ 40,760	\$ 34,954

The following table presents inventory by reportable segment (in millions):

	December 31, 2024	December 31, 2023
Automotive	\$ 9,988	\$ 11,139
Energy generation and storage	2,029	2,487
Total	<u>\$ 12,017</u>	<u>\$ 13,626</u>

Note 18 - Restructuring and Other

In the second quarter of 2024, we initiated and substantially completed certain restructuring actions to reduce costs and improve efficiency. As a result, we recognized \$583 million of employee termination expenses in Restructuring and other in our consolidated income statement. These expenses were substantially paid with an immaterial accrual remaining in Accrued liabilities and other in our consolidated balance sheet as of December 31, 2024.

During the year ended December 31, 2022, we recorded an impairment loss of \$204 million as well as realized gains of \$64 million in connection with converting our holdings of digital assets into fiat currency. We also recorded other expenses of \$36 million during the second quarter of the year ended December 31, 2022, related to employee terminations.

ITEM 9. CHANGES IN AND DISAGREEMENTS WITH ACCOUNTANTS ON ACCOUNTING AND FINANCIAL DISCLOSURE

None.

ITEM 9A. CONTROLS AND PROCEDURES

Evaluation of Disclosure Controls and Procedures

Our management, with the participation of our Chief Executive Officer and our Chief Financial Officer, evaluated the effectiveness of our disclosure controls and procedures pursuant to Rule 13a-15 under the Securities Exchange Act of 1934, as amended (the "Exchange Act"). In designing and evaluating the disclosure controls and procedures, our management recognizes that any controls and procedures, no matter how well designed and operated, can provide only reasonable assurance of achieving the desired control objectives. In addition, the design of disclosure controls and procedures must reflect the fact that there are resource constraints and that our management is required to apply its judgment in evaluating the benefits of possible controls and procedures relative to their costs.

Based on this evaluation, our Chief Executive Officer and our Chief Financial Officer concluded that, as of December 31, 2024, our disclosure controls and procedures were designed at a reasonable assurance level and were effective to provide reasonable assurance that the information we are required to disclose in reports that we file or submit under the Exchange Act is recorded, processed, summarized and reported within the time periods specified in the SEC rules and forms, and that such information is accumulated and communicated to our management, including our Chief Executive Officer and our Chief Financial Officer, as appropriate, to allow timely decisions regarding required disclosures.

Management's Report on Internal Control over Financial Reporting

Our management is responsible for establishing and maintaining adequate internal control over financial reporting. Internal control over financial reporting is a process designed by, or under the supervision of, our Chief Executive Officer and Chief Financial Officer to provide reasonable assurance regarding the reliability of financial reporting and the preparation of financial statements for external purposes in accordance with generally accepted accounting principles and includes those policies and procedures that (1) pertain to the maintenance of records that in reasonable detail accurately and fairly reflect the transactions and dispositions of our assets; (2) provide reasonable assurance that transactions are recorded as necessary to permit preparation of financial statements in accordance with generally accepted accounting principles, and that our receipts and expenditures are being made only in accordance with authorizations of our management and directors and (3) provide reasonable assurance regarding prevention or timely detection of unauthorized acquisition, use or disposition of our assets that could have a material effect on the financial statements.

PART IV

ITEM 15. EXHIBITS AND FINANCIAL STATEMENT SCHEDULES

1. Financial statements (see *Index to Consolidated Financial Statements* in Part II, Item 8 of this report)
2. All financial statement schedules have been omitted since the required information was not applicable or was not present in amounts sufficient to require submission of the schedules, or because the information required is included in the consolidated financial statements or the accompanying notes
3. The exhibits listed in the following *Index to Exhibits* are filed or incorporated by reference as part of this report

INDEX TO EXHIBITS

Exhibit Number	Exhibit Description	Incorporated by Reference				Filed Herewith
		Form	File No.	Exhibit	Filing Date	
3.1	<u>Certificate of Formation of the Registrant</u>	10-Q	001-34756	3.1	July 24, 2024	
3.2	<u>Bylaws of the Registrant.</u>	10-Q	001-34756	3.2	July 24, 2024	
4.1	<u>Specimen common stock certificate of the Registrant.</u>	—	—	—	—	X
4.2	<u>Fifth Amended and Restated Investors' Rights Agreement, dated as of August 31, 2009, between Registrant and certain holders of the Registrant's capital stock named therein.</u>	S-1	333-164593	4.2	January 29, 2010	
4.3	<u>Amendment to Fifth Amended and Restated Investors' Rights Agreement, dated as of May 20, 2010, between Registrant and certain holders of the Registrant's capital stock named therein.</u>	S-1/A	333-164593	4.2A	May 27, 2010	
4.4	<u>Amendment to Fifth Amended and Restated Investors' Rights Agreement between Registrant, Toyota Motor Corporation and certain holders of the Registrant's capital stock named therein.</u>	S-1/A	333-164593	4.2B	May 27, 2010	
4.5	<u>Amendment to Fifth Amended and Restated Investor's Rights Agreement, dated as of June 14, 2010, between Registrant and certain holders of the Registrant's capital stock named therein.</u>	S-1/A	333-164593	4.2C	June 15, 2010	
4.6	<u>Amendment to Fifth Amended and Restated Investor's Rights Agreement, dated as of November 2, 2010, between Registrant and certain holders of the Registrant's capital stock named therein.</u>	8-K	001-34756	4.1	November 4, 2010	
4.7	<u>Waiver to Fifth Amended and Restated Investor's Rights Agreement, dated as of May 22, 2011, between Registrant and certain holders of the Registrant's capital stock named therein.</u>	S-1/A	333-174466	4.2E	June 2, 2011	

Exhibit Number	Exhibit Description	Incorporated by Reference				Filed Herewith
		Form	File No.	Exhibit	Filing Date	
4.8	<u>Amendment to Fifth Amended and Restated Investor's Rights Agreement, dated as of May 30, 2011, between Registrant and certain holders of the Registrant's capital stock named therein.</u>	8-K	001-34756	4.1	June 1, 2011	
4.9	<u>Sixth Amendment to Fifth Amended and Restated Investors' Rights Agreement, dated as of May 15, 2013 among the Registrant, the Elon Musk Revocable Trust dated July 22, 2003 and certain other holders of the capital stock of the Registrant named therein.</u>	8-K	001-34756	4.1	May 20, 2013	
4.10	<u>Waiver to Fifth Amended and Restated Investor's Rights Agreement, dated as of May 14, 2013, between the Registrant and certain holders of the capital stock of the Registrant named therein.</u>	8-K	001-34756	4.2	May 20, 2013	
4.11	<u>Waiver to Fifth Amended and Restated Investor's Rights Agreement, dated as of August 13, 2015, between the Registrant and certain holders of the capital stock of the Registrant named therein.</u>	8-K	001-34756	4.1	August 19, 2015	
4.12	<u>Waiver to Fifth Amended and Restated Investors' Rights Agreement, dated as of May 18, 2016, between the Registrant and certain holders of the capital stock of the Registrant named therein.</u>	8-K	001-34756	4.1	May 24, 2016	
4.13	<u>Waiver to Fifth Amended and Restated Investors' Rights Agreement, dated as of March 15, 2017, between the Registrant and certain holders of the capital stock of the Registrant named therein.</u>	8-K	001-34756	4.1	March 17, 2017	
4.14	<u>Waiver to Fifth Amended and Restated Investors' Rights Agreement, dated as of May 1, 2019, between the Registrant and certain holders of the capital stock of the Registrant named therein.</u>	8-K	001-34756	4.1	May 3, 2019	
4.15	<u>Indenture, dated as of May 22, 2013, by and between the Registrant and U.S. Bank National Association.</u>	8-K	001-34756	4.1	May 22, 2013	
4.16	<u>Indenture, dated as of October 15, 2014, between SolarCity and U.S. Bank National Association, as trustee.</u>	S-3ASR(1)	333-199321	4.1	October 15, 2014	
4.17	<u>Tenth Supplemental Indenture, dated as of March 9, 2015, by and between SolarCity and the Trustee, related to SolarCity's 5.00% Solar Bonds, Series 2015/6-10.</u>	8-K(1)	001-35758	4.3	March 9, 2015	

Exhibit Number	Exhibit Description	Incorporated by Reference				Filed Herewith
		Form	File No.	Exhibit	Filing Date	
4.18	<u>Eleventh Supplemental Indenture, dated as of March 9, 2015, by and between SolarCity and the Trustee, related to SolarCity's 5.75% Solar Bonds, Series 2015/7-15.</u>	8-K(1)	001-35758	4.4	March 9, 2015	
4.19	<u>Fifteenth Supplemental Indenture, dated as of March 19, 2015, by and between SolarCity and the Trustee, related to SolarCity's 4.70% Solar Bonds, Series 2015/C4-10.</u>	8-K(1)	001-35758	4.5	March 19, 2015	
4.20	<u>Sixteenth Supplemental Indenture, dated as of March 19, 2015, by and between SolarCity and the Trustee, related to SolarCity's 5.45% Solar Bonds, Series 2015/C5-15.</u>	8-K(1)	001-35758	4.6	March 19, 2015	
4.21	<u>Twentieth Supplemental Indenture, dated as of March 26, 2015, by and between SolarCity and the Trustee, related to SolarCity's 4.70% Solar Bonds, Series 2015/C9-10.</u>	8-K(1)	001-35758	4.5	March 26, 2015	
4.22	<u>Twenty-First Supplemental Indenture, dated as of March 26, 2015, by and between SolarCity and the Trustee, related to SolarCity's 5.45% Solar Bonds, Series 2015/C10-15.</u>	8-K(1)	001-35758	4.6	March 26, 2015	
4.23	<u>Twenty-Sixth Supplemental Indenture, dated as of April 2, 2015, by and between SolarCity and the Trustee, related to SolarCity's 4.70% Solar Bonds, Series 2015/C14-10.</u>	8-K(1)	001-35758	4.5	April 2, 2015	
4.24	<u>Thirtieth Supplemental Indenture, dated as of April 9, 2015, by and between SolarCity and the Trustee, related to SolarCity's 4.70% Solar Bonds, Series 2015/C19-10.</u>	8-K(1)	001-35758	4.5	April 9, 2015	
4.25	<u>Thirty-First Supplemental Indenture, dated as of April 9, 2015, by and between SolarCity and the Trustee, related to SolarCity's 5.45% Solar Bonds, Series 2015/C20-15.</u>	8-K(1)	001-35758	4.6	April 9, 2015	
4.26	<u>Thirty-Fifth Supplemental Indenture, dated as of April 14, 2015, by and between SolarCity and the Trustee, related to SolarCity's 4.70% Solar Bonds, Series 2015/C24-10.</u>	8-K(1)	001-35758	4.5	April 14, 2015	
4.27	<u>Thirty-Sixth Supplemental Indenture, dated as of April 14, 2015, by and between SolarCity and the Trustee, related to SolarCity's 5.45% Solar Bonds, Series 2015/C25-15.</u>	8-K(1)	001-35758	4.6	April 14, 2015	

Exhibit Number	Exhibit Description	Incorporated by Reference				Filed Herewith
		Form	File No.	Exhibit	Filing Date	
4.28	<u>Thirty-Eighth Supplemental Indenture, dated as of April 21, 2015, by and between SolarCity and the Trustee, related to SolarCity's 4.70% Solar Bonds, Series 2015/C27-10.</u>	8-K(1)	001-35758	4.3	April 21, 2015	
4.29	<u>Thirty-Ninth Supplemental Indenture, dated as of April 21, 2015, by and between SolarCity and the Trustee, related to SolarCity's 5.45% Solar Bonds, Series 2015/C28-15.</u>	8-K(1)	001-35758	4.4	April 21, 2015	
4.30	<u>Forty-Third Supplemental Indenture, dated as of April 27, 2015, by and between SolarCity and the Trustee, related to SolarCity's 4.70% Solar Bonds, Series 2015/C32-10.</u>	8-K(1)	001-35758	4.5	April 27, 2015	
4.31	<u>Forty-Fourth Supplemental Indenture, dated as of April 27, 2015, by and between SolarCity and the Trustee, related to SolarCity's 5.45% Solar Bonds, Series 2015/C33-15.</u>	8-K(1)	001-35758	4.6	April 27, 2015	
4.32	<u>Forty-Eighth Supplemental Indenture, dated as of May 1, 2015, by and between SolarCity and the Trustee, related to SolarCity's 5.00% Solar Bonds, Series 2015/12-10.</u>	8-K(1)	001-35758	4.5	May 1, 2015	
4.33	<u>Forty-Ninth Supplemental Indenture, dated as of May 1, 2015, by and between SolarCity and the Trustee, related to SolarCity's 5.75% Solar Bonds, Series 2015/13-15.</u>	8-K(1)	001-35758	4.6	May 1, 2015	
4.34	<u>Fifty-Second Supplemental Indenture, dated as of May 11, 2015, by and between SolarCity and the Trustee, related to SolarCity's 4.70% Solar Bonds, Series 2015/C36-10.</u>	8-K(1)	001-35758	4.4	May 11, 2015	
4.35	<u>Fifty-Third Supplemental Indenture, dated as of May 11, 2015, by and between SolarCity and the Trustee, related to SolarCity's 5.45% Solar Bonds, Series 2015/C37-15.</u>	8-K(1)	001-35758	4.5	May 11, 2015	
4.36	<u>Fifty-Seventh Supplemental Indenture, dated as of May 18, 2015, by and between SolarCity and the Trustee, related to SolarCity's 4.70% Solar Bonds, Series 2015/C40-10.</u>	8-K(1)	001-35758	4.4	May 18, 2015	
4.37	<u>Fifty-Eighth Supplemental Indenture, dated as of May 18, 2015, by and between SolarCity and the Trustee, related to SolarCity's 5.45% Solar Bonds, Series 2015/C41-15.</u>	8-K(1)	001-35758	4.5	May 18, 2015	

Exhibit Number	Exhibit Description	Incorporated by Reference				Filed Herewith
		Form	File No.	Exhibit	Filing Date	
4.38	<u>Sixty-First Supplemental Indenture, dated as of May 26, 2015, by and between SolarCity and the Trustee, related to SolarCity's 4.70% Solar Bonds, Series 2015/C44-10.</u>	8-K(1)	001-35758	4.4	May 26, 2015	
4.39	<u>Sixty-Second Supplemental Indenture, dated as of May 26, 2015, by and between SolarCity and the Trustee, related to SolarCity's 5.45% Solar Bonds, Series 2015/C45-15.</u>	8-K(1)	001-35758	4.5	May 26, 2015	
4.40	<u>Seventieth Supplemental Indenture, dated as of June 16, 2015, by and between SolarCity and the Trustee, related to SolarCity's 4.70% Solar Bonds, Series 2015/C52-10.</u>	8-K(1)	001-35758	4.4	June 16, 2015	
4.41	<u>Seventy-First Supplemental Indenture, dated as of June 16, 2015, by and between SolarCity and the Trustee, related to SolarCity's 5.45% Solar Bonds, Series 2015/C53-15.</u>	8-K(1)	001-35758	4.5	June 16, 2015	
4.42	<u>Seventy-Fourth Supplemental Indenture, dated as of June 22, 2015, by and between SolarCity and the Trustee, related to SolarCity's 4.70% Solar Bonds, Series 2015/C56-10.</u>	8-K(1)	001-35758	4.4	June 23, 2015	
4.43	<u>Seventy-Fifth Supplemental Indenture, dated as of June 22, 2015, by and between SolarCity and the Trustee, related to SolarCity's 5.45% Solar Bonds, Series 2015/C57-15.</u>	8-K(1)	001-35758	4.5	June 23, 2015	
4.44	<u>Eightieth Supplemental Indenture, dated as of June 29, 2015, by and between SolarCity and the Trustee, related to SolarCity's 4.70% Solar Bonds, Series 2015/C61-10.</u>	8-K(1)	001-35758	4.5	June 29, 2015	
4.45	<u>Eighty-First Supplemental Indenture, dated as of June 29, 2015, by and between SolarCity and the Trustee, related to SolarCity's 5.45% Solar Bonds, Series 2015/C62-15.</u>	8-K(1)	001-35758	4.6	June 29, 2015	
4.46	<u>Ninetieth Supplemental Indenture, dated as of July 20, 2015, by and between SolarCity and the Trustee, related to SolarCity's 4.70% Solar Bonds, Series 2015/C71-10.</u>	8-K(1)	001-35758	4.5	July 21, 2015	
4.47	<u>Ninety-First Supplemental Indenture, dated as of July 20, 2015, by and between SolarCity and the Trustee, related to SolarCity's 5.45% Solar Bonds, Series 2015/C72-15.</u>	8-K(1)	001-35758	4.6	July 21, 2015	

Exhibit Number	Exhibit Description	Incorporated by Reference				Filed Herewith
		Form	File No.	Exhibit	Filing Date	
4.48	<u>Ninety-Fifth Supplemental Indenture, dated as of July 31, 2015, by and between SolarCity and the Trustee, related to SolarCity's 5.00% Solar Bonds, Series 2015/20-10.</u>	8-K(1)	001-35758	4.5	July 31, 2015	
4.49	<u>Ninety-Sixth Supplemental Indenture, dated as of July 31, 2015, by and between SolarCity and the Trustee, related to SolarCity's 5.75% Solar Bonds, Series 2015/21-15.</u>	8-K(1)	001-35758	4.6	July 31, 2015	
4.50	<u>One Hundred-and-Fifth Supplemental Indenture, dated as of August 10, 2015, by and between SolarCity and the Trustee, related to SolarCity's 4.70% Solar Bonds, Series 2015/C81-10.</u>	8-K(1)	001-35758	4.5	August 10, 2015	
4.51	<u>One Hundred-and-Eleventh Supplemental Indenture, dated as of August 17, 2015, by and between SolarCity and the Trustee, related to SolarCity's 5.45% Solar Bonds, Series 2015/C87-15.</u>	8-K(1)	001-35758	4.6	August 17, 2015	
4.52	<u>One Hundred-and-Sixteenth Supplemental Indenture, dated as of August 24, 2015, by and between SolarCity and the Trustee, related to SolarCity's 5.45% Solar Bonds, Series 2015/C92-15.</u>	8-K(1)	001-35758	4.6	August 24, 2015	
4.53	<u>One Hundred-and-Twenty-First Supplemental Indenture, dated as of August 31, 2015, by and between SolarCity and the Trustee, related to SolarCity's 5.45% Solar Bonds, Series 2015/C97-15.</u>	8-K(1)	001-35758	4.6	August 31, 2015	
4.54	<u>One Hundred-and-Twenty-Eighth Supplemental Indenture, dated as of September 14, 2015, by and between SolarCity and the Trustee, related to SolarCity's 4.70% Solar Bonds, Series 2015/C101-10.</u>	8-K(1)	001-35758	4.5	September 15, 2015	
4.55	<u>One Hundred-and-Twenty-Ninth Supplemental Indenture, dated as of September 14, 2015, by and between SolarCity and the Trustee, related to SolarCity's 5.45% Solar Bonds, Series 2015/C102-15.</u>	8-K(1)	001-35758	4.6	September 15, 2015	
4.56	<u>One Hundred-and-Thirty-Third Supplemental Indenture, dated as of September 28, 2015, by and between SolarCity and the Trustee, related to SolarCity's 4.70% Solar Bonds, Series 2015/C106-10.</u>	8-K(1)	001-35758	4.5	September 29, 2015	

Exhibit Number	Exhibit Description	Incorporated by Reference				Filed Herewith
		Form	File No.	Exhibit	Filing Date	
4.57	<u>One Hundred-and-Thirty-Fourth Supplemental Indenture, dated as of September 28, 2015, by and between SolarCity and the Trustee, related to SolarCity's 5.45% Solar Bonds, Series 2015/C107-15.</u>	8-K(1)	001-35758	4.6	September 29, 2015	
4.58	<u>One Hundred-and-Thirty-Eighth Supplemental Indenture, dated as of October 13, 2015, by and between SolarCity and the Trustee, related to SolarCity's 4.70% Solar Bonds, Series 2015/C111-10.</u>	8-K(1)	001-35758	4.5	October 13, 2015	
4.59	<u>One Hundred-and-Forty-Third Supplemental Indenture, dated as of October 30, 2015, by and between SolarCity and the Trustee, related to SolarCity's 5.00% Solar Bonds, Series 2015/25-10.</u>	8-K(1)	001-35758	4.5	October 30, 2015	
4.60	<u>One Hundred-and-Forty-Fourth Supplemental Indenture, dated as of October 30, 2015, by and between SolarCity and the Trustee, related to SolarCity's 5.75% Solar Bonds, Series 2015/26-15.</u>	8-K(1)	001-35758	4.6	October 30, 2015	
4.61	<u>One Hundred-and-Forty-Eighth Supplemental Indenture, dated as of November 4, 2015, by and between SolarCity and the Trustee, related to SolarCity's 4.70% Solar Bonds, Series 2015/C116-10.</u>	8-K(1)	001-35758	4.5	November 4, 2015	
4.62	<u>One Hundred-and-Fifty-Third Supplemental Indenture, dated as of November 16, 2015, by and between SolarCity and the Trustee, related to SolarCity's 4.70% Solar Bonds, Series 2015/C121-10.</u>	8-K(1)	001-35758	4.5	November 17, 2015	
4.63	<u>One Hundred-and-Fifty-Fourth Supplemental Indenture, dated as of November 16, 2015, by and between SolarCity and the Trustee, related to SolarCity's 5.45% Solar Bonds, Series 2015/C122-15.</u>	8-K(1)	001-35758	4.6	November 17, 2015	
4.64	<u>One Hundred-and-Fifty-Eighth Supplemental Indenture, dated as of November 30, 2015, by and between SolarCity and the Trustee, related to SolarCity's 4.70% Solar Bonds, Series 2015/C126-10.</u>	8-K(1)	001-35758	4.5	November 30, 2015	
4.65	<u>One Hundred-and-Fifty-Ninth Supplemental Indenture, dated as of November 30, 2015, by and between SolarCity and the Trustee, related to SolarCity's 5.45% Solar Bonds, Series 2015/C127-15.</u>	8-K(1)	001-35758	4.6	November 30, 2015	

Exhibit Number	Exhibit Description	Incorporated by Reference				Filed Herewith
		Form	File No.	Exhibit	Filing Date	
4.66	<u>One Hundred-and-Sixty-Third Supplemental Indenture, dated as of December 14, 2015, by and between SolarCity and the Trustee, related to SolarCity's 4.70% Solar Bonds, Series 2015/C131-10.</u>	8-K(1)	001-35758	4.5	December 14, 2015	
4.67	<u>One Hundred-and-Sixty-Fourth Supplemental Indenture, dated as of December 14, 2015, by and between SolarCity and the Trustee, related to SolarCity's 5.45% Solar Bonds, Series 2015/C132-15.</u>	8-K(1)	001-35758	4.6	December 14, 2015	
4.68	<u>One Hundred-and-Sixty-Eighth Supplemental Indenture, dated as of December 28, 2015, by and between SolarCity and the Trustee, related to SolarCity's 4.70% Solar Bonds, Series 2015/C136-10.</u>	8-K(1)	001-35758	4.5	December 28, 2015	
4.69	<u>One Hundred-and-Sixty-Ninth Supplemental Indenture, dated as of December 28, 2015, by and between SolarCity and the Trustee, related to SolarCity's 5.45% Solar Bonds, Series 2015/C137-15.</u>	8-K(1)	001-35758	4.6	December 28, 2015	
4.70	<u>One Hundred-and-Seventy-Third Supplemental Indenture, dated as of January 29, 2016, by and between SolarCity and the Trustee, related to SolarCity's 5.00% Solar Bonds, Series 2016/4-10.</u>	8-K(1)	001-35758	4.5	January 29, 2016	
4.71	<u>One Hundred-and-Seventy-Fourth Supplemental Indenture, dated as of January 29, 2016, by and between SolarCity and the Trustee, related to SolarCity's 5.75% Solar Bonds, Series 2016/5-15.</u>	8-K(1)	001-35758	4.6	January 29, 2016	
4.72	<u>Description of Registrant's Securities</u>	—	—	—	—	X
10.1**	<u>Form of Indemnification Agreement between the Registrant and its directors and officers.</u>	S-1/A	333-164593	10.1	June 15, 2010	
10.2**	<u>2003 Equity Incentive Plan.</u>	S-1/A	333-164593	10.2	May 27, 2010	
10.3**	<u>Form of Stock Option Agreement under 2003 Equity Incentive Plan.</u>	S-1	333-164593	10.3	January 29, 2010	
10.4**	<u>Amended and Restated 2010 Equity Incentive Plan.</u>	10-K	001-34756	10.4	February 23, 2018	
10.5**	<u>Form of Stock Option Agreement under 2010 Equity Incentive Plan.</u>	10-K	001-34756	10.6	March 1, 2017	
10.6**	<u>Form of Restricted Stock Unit Award Agreement under 2010 Equity Incentive Plan.</u>	10-K	001-34756	10.7	March 1, 2017	

Exhibit Number	Exhibit Description	Incorporated by Reference				Filed Herewith
		Form	File No.	Exhibit	Filing Date	
10.7**	<u>Amended and Restated 2010 Employee Stock Purchase Plan, effective as of February 1, 2017.</u>	10-K	001-34756	10.8	March 1, 2017	
10.8**	<u>2019 Equity Incentive Plan.</u>	S-8	333-232079	4.2	June 12, 2019	
10.9**	<u>Form of Stock Option Agreement under 2019 Equity Incentive Plan.</u>	S-8	333-232079	4.3	June 12, 2019	
10.10**	<u>Form of Restricted Stock Unit Award Agreement under 2019 Equity Incentive Plan.</u>	S-8	333-232079	4.4	June 12, 2019	
10.11**	<u>Employee Stock Purchase Plan, effective as of June 12, 2019.</u>	S-8	333-232079	4.5	June 12, 2019	
10.12**	<u>2012 SolarCity Equity Incentive Plan and form of agreements used thereunder.</u>	S-1(1)	333-184317	10.3	October 5, 2012	
10.13**	<u>Offer Letter between the Registrant and Elon Musk dated October 13, 2008.</u>	S-1	333-164593	10.9	January 29, 2010	
10.14**	<u>Performance Stock Option Agreement between the Registrant and Elon Musk dated January 21, 2018.</u>	DEF 14A	001-34756	Appendix A	February 8, 2018	
10.15	<u>Indemnification Agreement, effective as of June 23, 2020, between Registrant and Elon R. Musk.</u>	10-Q	001-34756	10.4	July 28, 2020	
10.16	<u>Form of Call Option Confirmation relating to 1.25% Convertible Senior Notes Due March 1, 2021.</u>	8-K	001-34756	10.3	March 5, 2014	
10.17†	<u>Supply Agreement between Panasonic Corporation and the Registrant dated October 5, 2011.</u>	10-K	001-34756	10.50	February 27, 2012	
10.18†	<u>Amendment No. 1 to Supply Agreement between Panasonic Corporation and the Registrant dated October 29, 2013.</u>	10-K	001-34756	10.35A	February 26, 2014	
10.19	<u>Agreement between Panasonic Corporation and the Registrant dated July 31, 2014.</u>	10-Q	001-34756	10.1	November 7, 2014	
10.20†	<u>General Terms and Conditions between Panasonic Corporation and the Registrant dated October 1, 2014.</u>	8-K	001-34756	10.2	October 11, 2016	
10.21	<u>Letter Agreement, dated as of February 24, 2015, regarding addition of co-party to General Terms and Conditions, Production Pricing Agreement and Investment Letter Agreement between Panasonic Corporation and the Registrant.</u>	10-K	001-34756	10.25A	February 24, 2016	

Exhibit Number	Exhibit Description	Incorporated by Reference				Filed Herewith
		Form	File No.	Exhibit	Filing Date	
10.22†	<u>Investment Letter Agreement between Panasonic Corporation and the Registrant dated October 1, 2014.</u>	10-Q	001-34756	10.4	November 7, 2014	
10.23	<u>Amendment to Gigafactory Documents, dated April 5, 2016, by and among the Registrant, Panasonic Corporation, Panasonic Corporation of North America and Panasonic Energy Corporation of North America.</u>	10-Q	001-34756	10.2	May 10, 2016	
10.24††	<u>2019 Pricing Agreement (Japan Cells) with respect to 2011 Supply Agreement, executed September 20, 2019, by and among the Registrant, Tesla Motors Netherlands B.V., Panasonic Corporation and SANYO Electric Co., Ltd.</u>	10-Q	001-34756	10.6	October 29, 2019	
10.25††	<u>2020 Pricing Agreement (Gigafactory 2170 Cells), entered into on June 9, 2020, by and among Registrant, Tesla Motors Netherlands B.V., Panasonic Corporation and Panasonic Corporation of North America.</u>	10-Q	001-34756	10.3	July 28, 2020	
10.26††	<u>2021 Pricing Agreement (Japan Cells) with respect to 2011 Supply Agreement, executed December 29, 2020, by and among the Registrant, Tesla Motors Netherlands B.V., Panasonic Corporation of North America and SANYO Electric Co., Ltd.</u>	10-K	001-34756	10.39	February 8, 2021	
10.27††	<u>Amended and Restated Factory Lease, executed as of March 26, 2019, by and between the Registrant and Panasonic Energy North America, a division of Panasonic Corporation of North America, as tenant.</u>	10-Q	001-34756	10.3	July 29, 2019	
10.28††	<u>Lease Amendment, executed September 20, 2019, by and among the Registrant, Panasonic Corporation of North America, on behalf of its division Panasonic Energy of North America, with respect to the Amended and Restated Factory Lease, executed as of March 26, 2019.</u>	10-Q	001-34756	10.7	October 29, 2019	
10.29††	<u>Second Lease Amendment, entered into on June 9, 2020, by and between the Registrant and Panasonic Energy of North America, a division of Panasonic Corporation of North America, with respect to the Amended and Restated Factory Lease dated January 1, 2017.</u>	10-Q	001-34756	10.1	July 28, 2020	
10.30†	<u>Agreement for Tax Abatement and Incentives, dated as of May 7, 2015, by and between Tesla Motors, Inc. and the State of Nevada, acting by and through the Nevada Governor's Office of Economic Development.</u>	10-Q	001-34756	10.1	August 7, 2015	

Exhibit Number	Exhibit Description	Incorporated by Reference				Filed Herewith
		Form	File No.	Exhibit	Filing Date	
10.31	<u>Amended and Restated Agreement For Research & Development Alliance on Triex Module Technology, effective as of September 2, 2014, by and between The Research Foundation For The State University of New York, on behalf of the College of Nanoscale Science and Engineering of the State University of New York, and Silevo, Inc.</u>	10-Q(1)	001-35758	10.16	November 6, 2014	
10.32	<u>First Amendment to Amended and Restated Agreement For Research & Development Alliance on Triex Module Technology, effective as of October 31, 2014, by and between The Research Foundation For The State University of New York, on behalf of the College of Nanoscale Science and Engineering of the State University of New York, and Silevo, Inc.</u>	10-K(1)	001-35758	10.16a	February 24, 2015	
10.33	<u>Second Amendment to Amended and Restated Agreement For Research & Development Alliance on Triex Module Technology, effective as of December 15, 2014, by and between The Research Foundation For The State University of New York, on behalf of the College of Nanoscale Science and Engineering of the State University of New York, and Silevo, Inc.</u>	10-K(1)	001-35758	10.16b	February 24, 2015	
10.34	<u>Third Amendment to Amended and Restated Agreement For Research & Development Alliance on Triex Module Technology, effective as of February 12, 2015, by and between The Research Foundation For The State University of New York, on behalf of the College of Nanoscale Science and Engineering of the State University of New York, and Silevo, Inc.</u>	10-Q(1)	001-35758	10.16c	May 6, 2015	
10.35	<u>Fourth Amendment to Amended and Restated Agreement For Research & Development Alliance on Triex Module Technology, effective as of March 30, 2015, by and between The Research Foundation For The State University of New York, on behalf of the College of Nanoscale Science and Engineering of the State University of New York, and Silevo, Inc.</u>	10-Q(1)	001-35758	10.16d	May 6, 2015	
10.36	<u>Fifth Amendment to Amended and Restated Agreement For Research & Development Alliance on Triex Module Technology, effective as of June 30, 2015, by and between The Research Foundation For The State University of New York, on behalf of the College of Nanoscale Science and Engineering of the State University of New York, and Silevo, LLC.</u>	10-Q(1)	001-35758	10.16e	July 30, 2015	

Exhibit Number	Exhibit Description	Incorporated by Reference				Filed Herewith
		Form	File No.	Exhibit	Filing Date	
10.37	<u>Sixth Amendment to Amended and Restated Agreement For Research & Development Alliance on Triex Module Technology, effective as of September 1, 2015, by and between The Research Foundation For The State University of New York, on behalf of the College of Nanoscale Science and Engineering of the State University of New York, and Silevo, LLC.</u>	10-Q(1)	001-35758	10.16f	October 30, 2015	
10.38	<u>Seventh Amendment to Amended and Restated Agreement For Research & Development Alliance on Triex Module Technology, effective as of October 9, 2015, by and between The Research Foundation For The State University of New York, on behalf of the College of Nanoscale Science and Engineering of the State University of New York, and Silevo, LLC.</u>	10-Q(1)	001-35758	10.16g	October 30, 2015	
10.39	<u>Eighth Amendment to Amended and Restated Agreement For Research & Development Alliance on Triex Module Technology, effective as of October 26, 2015, by and between The Research Foundation For The State University of New York, on behalf of the College of Nanoscale Science and Engineering of the State University of New York, and Silevo, LLC.</u>	10-Q(1)	001-35758	10.16h	October 30, 2015	
10.40	<u>Ninth Amendment to Amended and Restated Agreement For Research & Development Alliance on Triex Module Technology, effective as of December 9, 2015, by and between The Research Foundation For The State University of New York, on behalf of the College of Nanoscale Science and Engineering of the State University of New York, and Silevo, LLC.</u>	10-K(1)	001-35758	10.16i	February 10, 2016	
10.41	<u>Tenth Amendment to Amended and Restated Agreement For Research & Development Alliance on Triex Module Technology, effective as of March 31, 2017, by and between The Research Foundation For The State University of New York, on behalf of the Colleges of Nanoscale Science and Engineering of the State University of New York, and Silevo, LLC.</u>	10-Q	001-34756	10.8	May 10, 2017	
10.42	<u>Eleventh Amendment to Amended and Restated Agreement for Research & Development Alliance on Triex Module Technology, effective as of July 22, 2020, among the Research Foundation for the State University of New York, Silevo, LLC and Tesla Energy Operations, Inc.</u>	10-Q	001-34756	10.6	July 28, 2020	

Exhibit Number	Exhibit Description	Incorporated by Reference				Filed Herewith
		Form	File No.	Exhibit	Filing Date	
10.43	<u>Twelfth Amendment to Amended and Restated Agreement for Research & Development Alliance on Triex Module Technology, effective as of May 1, 2021, among the Research Foundation for the State University of New York, Silevo, LLC and Tesla Energy Operations, Inc.</u>	10-Q	001-34756	10.1	October 25, 2021	
10.44†	<u>Grant Contract for State-Owned Construction Land Use Right, dated as of October 17, 2018, by and between Shanghai Planning and Land Resource Administration Bureau, as grantor, and Tesla (Shanghai) Co., Ltd., as grantee (English translation).</u>	10-Q	001-34756	10.2	July 29, 2019	
10.45	<u>Credit Agreement, dated as of January 20, 2023, among Tesla, Inc., the Lenders and Issuing Banks from time to time party thereto, Citibank, N.A., as Administrative Agent and Deutsche Bank Securities, Inc., as Syndication Agent</u>	10-K	001-34756	10.59	January 31, 2023	
19	<u>Insider Trading Policy</u>	—	—	—	—	X
21.1	<u>List of Subsidiaries of the Registrant</u>	—	—	—	—	X
23.1	<u>Consent of PricewaterhouseCoopers LLP, Independent Registered Public Accounting Firm</u>	—	—	—	—	X
31.1	<u>Rule 13a-14(a) / 15(d)-14(a) Certification of Principal Executive Officer</u>	—	—	—	—	X
31.2	<u>Rule 13a-14(a) / 15(d)-14(a) Certification of Principal Financial Officer</u>	—	—	—	—	X
32.1*	<u>Section 1350 Certifications</u>	—	—	—	—	X
97	<u>Tesla, Inc. Clawback Policy</u>	10-K	001-34756	97	January 29, 2024	
101.INS	Inline XBRL Instance Document	—	—	—	—	X
101.SCH	Inline XBRL Taxonomy Extension Schema Document	—	—	—	—	X
101.CAL	Inline XBRL Taxonomy Extension Calculation Linkbase Document.	—	—	—	—	X
101.DEF	Inline XBRL Taxonomy Extension Definition Linkbase Document	—	—	—	—	X
101.LAB	Inline XBRL Taxonomy Extension Label Linkbase Document	—	—	—	—	X

Exhibit Number	Exhibit Description	Incorporated by Reference				Filed Herewith
		Form	File No.	Exhibit	Filing Date	
101.PRE	Inline XBRL Taxonomy Extension Presentation Linkbase Document	—	—	—	—	X
104	Cover Page Interactive Data File (formatted as inline XBRL with applicable taxonomy extension information contained in Exhibits 101)					

* Furnished herewith

** Indicates a management contract or compensatory plan or arrangement

† Confidential treatment has been requested for portions of this exhibit

†† Portions of this exhibit have been redacted in compliance with Regulation S-K Item 601(b)(10).

(1) Indicates a filing of SolarCity

(2) Indicates a filing of Maxwell Technologies, Inc.

ITEM 16. FORM 10-K SUMMARY

None.

SIGNATURES

Pursuant to the requirements of Section 13 or 15(d) the Securities Exchange Act of 1934, the registrant has duly caused this report to be signed on its behalf by the undersigned, thereunto duly authorized.

Tesla, Inc.

Date: January 29, 2025

/s/ Elon Musk

Elon Musk

Chief Executive Officer
(Principal Executive Officer)

Pursuant to the requirements of the Securities Exchange Act of 1934, this report has been signed below by the following persons on behalf of the registrant and in the capacities and on the dates indicated.

Signature	Title	Date
/s/ Elon Musk Elon Musk	Chief Executive Officer and Director (Principal Executive Officer)	January 29, 2025
/s/ Vaibhav Taneja Vaibhav Taneja	Chief Financial Officer (Principal Financial Officer and Principal Accounting Officer)	January 29, 2025
/s/ Robyn Denholm Robyn Denholm	Director	January 29, 2025
/s/ Ira Ehrenpreis Ira Ehrenpreis	Director	January 29, 2025
/s/ Joseph Gebbia Joseph Gebbia	Director	January 29, 2025
/s/ James Murdoch James Murdoch	Director	January 29, 2025
/s/ Kimbal Musk Kimbal Musk	Director	January 29, 2025
/s/ JB Straubel JB Straubel	Director	January 29, 2025
/s/ Kathleen Wilson-Thompson Kathleen Wilson-Thompson	Director	January 29, 2025